Preliminary Statistical Testing of SPFL Dataset

2026-07-18

Contents

Introduction

The aim is to conduct a preliminary statistical investigation into home advantage in the SPFL. The analysis explores whether home teams win or score more often, whether attendance and away-team travel distance are associated with home wins, and whether disciplinary patterns differ between home and away teams. These tests are exploratory and are used to identify patterns rather than prove causation.

Load packages and datasets

```r
# Load the packages required for importing Excel files, data manipulation,
# and producing plots.
library(readxl)
library(dplyr)
library(ggplot2)

# Define a function to load one season of SPFL data.
# The function reads an Excel file, optionally from a specified sheet,
# and adds a Season column so that each dataset can be identified later.
load_season <- function(file, season, sheet = NULL) {

  # If no sheet name is supplied, read the first sheet in the Excel file.
  if (is.null(sheet)) {
    data <- read_excel(file)
  } else {
    # If a sheet name is supplied, read that specific sheet.
    data <- read_excel(file, sheet = sheet)
  }

  # Add a Season column to the dataset.
  data %>%
    mutate(Season = season)
}

# Load each individual SPFL Premiership season dataset.
# The sheet argument is included where the relevant Excel file contains
# a named worksheet that needs to be specified.
spfl_2016_2017 <- load_season("SPFL 2016-2017.xlsx", "2016/17", "SPFL 2016-2017")
spfl_2017_2018 <- load_season("SPFL 2017-2018.xlsx", "2017/18", "SPFL 2017-2018")
spfl_2018_2019 <- load_season("SPFL 2018-2019.xlsx", "2018/19", "SPFL 2018-2019")
spfl_2019_2020 <- load_season("SPFL 2019-2020.xlsx", "2019/20", "SPFL 2019-2020")
spfl_2020_2021 <- load_season("SPFL 2020-2021.xlsx", "2020/21", "SPFL 2020-2021")
spfl_2021_2022 <- load_season("SPFL 2021-2022.xlsx", "2021/22")
spfl_2022_2023 <- load_season("SPFL 2022-2023.xlsx", "2022/23")
spfl_2023_2024 <- load_season("SPFL 2023-2024.xlsx", "2023/24")
spfl_2024_2025 <- load_season("SPFL 2024-2025.xlsx", "2024/25")
spfl_2025_2026 <- load_season("SPFL 2025-2026.xlsx", "2025/26", "SPFL Dataset")
```

```r
# Store all ten season datasets in a named list.
# This makes it easier to apply the same analysis to each season separately.
season_list <- list(
  "2016/17" = spfl_2016_2017,
  "2017/18" = spfl_2017_2018,
  "2018/19" = spfl_2018_2019,
  "2019/20" = spfl_2019_2020,
  "2020/21" = spfl_2020_2021,
  "2021/22" = spfl_2021_2022,
  "2022/23" = spfl_2022_2023,
  "2023/24" = spfl_2023_2024,
  "2024/25" = spfl_2024_2025,
  "2025/26" = spfl_2025_2026
)
```

Data cleaning/prep

```r
column_names_table <- bind_rows(
  lapply(names(season_list), function(season_name) {
    data.frame(
      Season = season_name,
      ColumnName = names(season_list[[season_name]])
    )
  })
)

column_names_table
```

```
##     Season                       ColumnName
## 1  2016/17                             Date
## 2  2016/17                        Home Team
## 3  2016/17                        Away Team
## 4  2016/17                       Home Goals
## 5  2016/17                       Away Goals
## 6  2016/17                           Result
## 7  2016/17                           Season
## 8  2017/18                             Date
## 9  2017/18                        Home Team
## 10 2017/18                        Away Team
## 11 2017/18                       Home Goals
## 12 2017/18                       Away Goals
## 13 2017/18                           Result
## 14 2017/18                           Season
## 15 2018/19                             Date
## 16 2018/19                        Home Team
## 17 2018/19                        Away Team
## 18 2018/19                       Home Goals
## 19 2018/19                       Away Goals
## 20 2018/19                           Result
## 21 2018/19                           Season
## 22 2019/20                             Date
## 23 2019/20                        Home Team
```

```
## 24 2019/20                      Away Team
## 25 2019/20                     Home Goals
## 26 2019/20                     Away Goals
## 27 2019/20                         Result
## 28 2019/20                         Season
## 29 2020/21                           Date
## 30 2020/21                      Home Team
## 31 2020/21                      Away Team
## 32 2020/21                     Home Goals
## 33 2020/21                     Away Goals
## 34 2020/21                         Result
## 35 2020/21                Goal Difference
## 36 2020/21              Home Yellow cards
## 37 2020/21                 Home Red cards
## 38 2020/21              Away Yellow cards
## 39 2020/21                 Away Red cards
## 40 2020/21                         Season
## 41 2021/22                           Date
## 42 2021/22                      Home Team
## 43 2021/22                      Away Team
## 44 2021/22                     Home Goals
## 45 2021/22                     Away Goals
## 46 2021/22                         Result
## 47 2021/22                         Season
## 48 2022/23                           Date
## 49 2022/23                      Home Team
## 50 2022/23                      Away Team
## 51 2022/23                     Home Goals
## 52 2022/23                     Away Goals
## 53 2022/23                         Result
## 54 2022/23                         Season
## 55 2023/24                           Date
## 56 2023/24                      Home Team
## 57 2023/24                      Away Team
## 58 2023/24                     Home Goals
## 59 2023/24                     Away Goals
## 60 2023/24                         Result
## 61 2023/24                         Season
## 62 2024/25                           Date
## 63 2024/25                      Home Team
## 64 2024/25                      Away Team
## 65 2024/25                     Home Goals
## 66 2024/25                     Away Goals
## 67 2024/25                         Result
## 68 2024/25             Overall Attendance
## 69 2024/25                          Teams
## 70 2024/25        Incorrect Decision for
## 71 2024/25    Incorrect Decision against
## 72 2024/25                         Season
## 73 2025/26                           Date
## 74 2025/26                      Home Team
## 75 2025/26                      Away Team
## 76 2025/26                     Home Goals
## 77 2025/26                     Away Goals
```

```
## 78 2025/26                              Result
## 79 2025/26                          Attendance
## 80 2025/26                    Percent Capacity
## 81 2025/26     Travel Distance for Away Team
## 82 2025/26         Travel time for Away Team
## 83 2025/26                               Teams
## 84 2025/26            Incorrect Decision for
## 85 2025/26        Incorrect Decision against
## 86 2025/26                           SPFL team
## 87 2025/26                    Stadium Capacity
## 88 2025/26                              Team A
## 89 2025/26                              Team B
## 90 2025/26 Distance via time Travelled (hrs)
## 91 2025/26  Distance between A and B (miles)
## 92 2025/26                     Goal Difference
## 93 2025/26                      Home Red cards
## 94 2025/26                   Home Yellow cards
## 95 2025/26                      Away Red cards
## 96 2025/26                   Away Yellow cards
## 97 2025/26                              Season
```

```r
unique_column_names <- sort(unique(unlist(
  lapply(season_list, names)
)))

unique_column_names
```

```
##  [1] "Attendance"                        "Away Goals"
##  [3] "Away Red cards"                    "Away Team"
##  [5] "Away Yellow cards"                 "Date"
##  [7] "Distance between A and B (miles)"  "Distance via time Travelled (hrs)"
##  [9] "Goal Difference"                   "Home Goals"
## [11] "Home Red cards"                    "Home Team"
## [13] "Home Yellow cards"                 "Incorrect Decision against"
## [15] "Incorrect Decision for"            "Overall Attendance"
## [17] "Percent Capacity"                  "Result"
## [19] "Season"                            "SPFL team"
## [21] "Stadium Capacity"                  "Team A"
## [23] "Team B"                            "Teams"
## [25] "Travel Distance for Away Team"     "Travel time for Away Team"
```

```r
team_names_table <- bind_rows(
  lapply(names(season_list), function(season_name) {

    teams <- sort(unique(c(
      season_list[[season_name]]$`Home Team`,
      season_list[[season_name]]$`Away Team`
    )))

    data.frame(
      Season = season_name,
      Team = teams
    )
  })
)
```

```
team_names_table
```

```
##      Season            Team
## 1   2016/17        Aberdeen
## 2   2016/17          Celtic
## 3   2016/17          Dundee
## 4   2016/17        Hamilton
## 5   2016/17          Hearts
## 6   2016/17    Inverness CT
## 7   2016/17      Kilmarnock
## 8   2016/17      Motherwell
## 9   2016/17 Partick Thistle
## 10  2016/17         Rangers
## 11  2016/17     Ross County
## 12  2016/17    St Johnstone
## 13  2017/18        Aberdeen
## 14  2017/18          Celtic
## 15  2017/18          Dundee
## 16  2017/18        Hamilton
## 17  2017/18          Hearts
## 18  2017/18       Hibernian
## 19  2017/18      Kilmarnock
## 20  2017/18      Motherwell
## 21  2017/18 Partick Thistle
## 22  2017/18         Rangers
## 23  2017/18     Ross County
## 24  2017/18    St Johnstone
## 25  2018/19        Aberdeen
## 26  2018/19          Celtic
## 27  2018/19          Dundee
## 28  2018/19        Hamilton
## 29  2018/19          Hearts
## 30  2018/19       Hibernian
## 31  2018/19      Kilmarnock
## 32  2018/19      Livingston
## 33  2018/19      Motherwell
## 34  2018/19         Rangers
## 35  2018/19    St Johnstone
## 36  2018/19       St Mirren
## 37  2019/20        Aberdeen
## 38  2019/20          Celtic
## 39  2019/20        Hamilton
## 40  2019/20          Hearts
## 41  2019/20       Hibernian
## 42  2019/20      Kilmarnock
## 43  2019/20      Livingston
## 44  2019/20      Motherwell
## 45  2019/20         Rangers
## 46  2019/20     Ross County
## 47  2019/20    St Johnstone
## 48  2019/20       St Mirren
## 49  2020/21        Aberdeen
## 50  2020/21          Celtic
```

```
## 51  2020/21   Dundee United
## 52  2020/21        Hamilton
## 53  2020/21       Hibernian
## 54  2020/21      Kilmarnock
## 55  2020/21      Livingston
## 56  2020/21      Motherwell
## 57  2020/21         Rangers
## 58  2020/21     Ross County
## 59  2020/21    St Johnstone
## 60  2020/21       St Mirren
## 61  2021/22        Aberdeen
## 62  2021/22          Celtic
## 63  2021/22          Dundee
## 64  2021/22   Dundee United
## 65  2021/22          Hearts
## 66  2021/22       Hibernian
## 67  2021/22      Livingston
## 68  2021/22      Motherwell
## 69  2021/22         Rangers
## 70  2021/22     Ross County
## 71  2021/22    St Johnstone
## 72  2021/22       St Mirren
## 73  2022/23        Aberdeen
## 74  2022/23          Celtic
## 75  2022/23   Dundee United
## 76  2022/23          Hearts
## 77  2022/23       Hibernian
## 78  2022/23      Kilmarnock
## 79  2022/23      Livingston
## 80  2022/23      Motherwell
## 81  2022/23         Rangers
## 82  2022/23     Ross County
## 83  2022/23    St Johnstone
## 84  2022/23       St Mirren
## 85  2023/24        Aberdeen
## 86  2023/24          Celtic
## 87  2023/24          Dundee
## 88  2023/24          Hearts
## 89  2023/24       Hibernian
## 90  2023/24      Kilmarnock
## 91  2023/24      Livingston
## 92  2023/24      Motherwell
## 93  2023/24         Rangers
## 94  2023/24     Ross County
## 95  2023/24    St Johnstone
## 96  2023/24       St Mirren
## 97  2024/25        Aberdeen
## 98  2024/25          Celtic
## 99  2024/25          Dundee
## 100 2024/25   Dundee United
## 101 2024/25          Hearts
## 102 2024/25       Hibernian
## 103 2024/25      Kilmarnock
## 104 2024/25      Motherwell
```

```
## 105 2024/25        Rangers
## 106 2024/25    Ross County
## 107 2024/25   St Johnstone
## 108 2024/25      St Mirren
## 109 2025/26       Aberdeen
## 110 2025/26         Celtic
## 111 2025/26         Dundee
## 112 2025/26  Dundee United
## 113 2025/26        Falkirk
## 114 2025/26         Hearts
## 115 2025/26      Hibernian
## 116 2025/26     Kilmarnock
## 117 2025/26     Livingston
## 118 2025/26     Motherwell
## 119 2025/26        Rangers
## 120 2025/26      St Mirren
```

```r
all_team_names <- sort(unique(team_names_table$Team))

all_team_names
```

```
##  [1] "Aberdeen"        "Celtic"          "Dundee"          "Dundee United"
##  [5] "Falkirk"         "Hamilton"        "Hearts"          "Hibernian"
##  [9] "Inverness CT"    "Kilmarnock"      "Livingston"      "Motherwell"
## [13] "Partick Thistle" "Rangers"         "Ross County"     "St Johnstone"
## [17] "St Mirren"
```

```r
teams_per_season <- team_names_table %>%
  group_by(Season) %>%
  summarise(
    NumberOfTeams = n_distinct(Team),
    Teams = paste(sort(unique(Team)), collapse = ", "),
    .groups = "drop"
  )

teams_per_season
```

```
## # A tibble: 10 x 3
##    Season  NumberOfTeams Teams
##    <chr>           <int> <chr>
##  1 2016/17            12 Aberdeen, Celtic, Dundee, Hamilton, Hearts, Inverness ~
##  2 2017/18            12 Aberdeen, Celtic, Dundee, Hamilton, Hearts, Hibernian,~
##  3 2018/19            12 Aberdeen, Celtic, Dundee, Hamilton, Hearts, Hibernian,~
##  4 2019/20            12 Aberdeen, Celtic, Hamilton, Hearts, Hibernian, Kilmarn~
##  5 2020/21            12 Aberdeen, Celtic, Dundee United, Hamilton, Hibernian, ~
##  6 2021/22            12 Aberdeen, Celtic, Dundee, Dundee United, Hearts, Hiber~
##  7 2022/23            12 Aberdeen, Celtic, Dundee United, Hearts, Hibernian, Ki~
##  8 2023/24            12 Aberdeen, Celtic, Dundee, Hearts, Hibernian, Kilmarnoc~
##  9 2024/25            12 Aberdeen, Celtic, Dundee, Dundee United, Hearts, Hiber~
## 10 2025/26            12 Aberdeen, Celtic, Dundee, Dundee United, Falkirk, Hear~
```

```r
options(scipen = 999)
```

Binomial test: are home wins more common than non-home wins?

A binomial test was used to test whether the proportion of home wins was significantly greater than 0.5. The null hypothesis was that home wins and non-home-wins were equally likely.

```
spfl <- spfl_2025_2026
home_wins <- sum(spfl$Result == "Home", na.rm = TRUE)
n_matches <- sum(!is.na(spfl$Result))

home_wins
```

```
## [1] 102
```

```
n_matches
```

```
## [1] 228
```

```
p_value <- sum(dbinom(home_wins:n_matches, n_matches, 0.5))
p_value
```

```
## [1] 0.951212
```

The p-value was approximately 0.95, which is greater than 0.05. Therefore, the null hypothesis cannot be rejected. There is insufficient evidence to suggest that home wins were more likely than non-home wins.

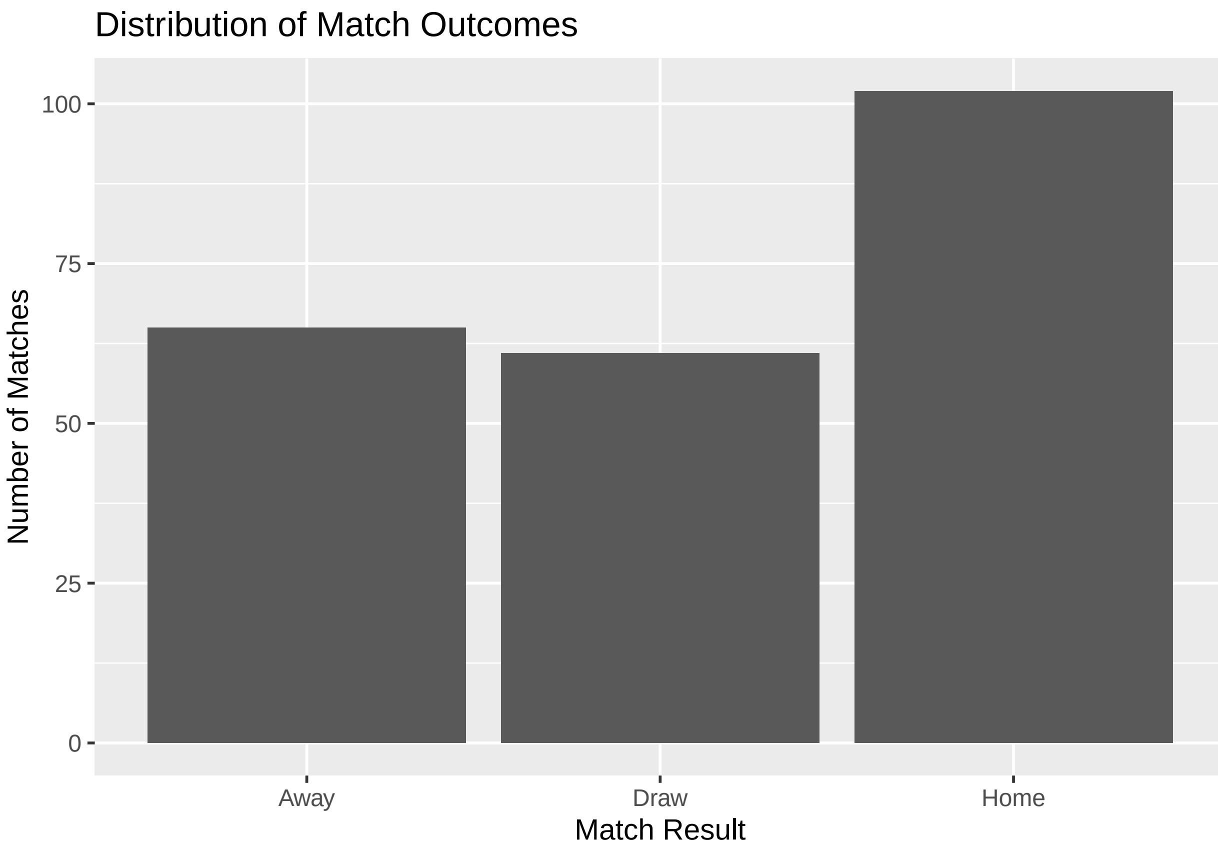
```
home_away_only <- spfl %>% filter(Result %in% c("Home", "Away"))

home_wins <- sum(home_away_only$Result == "Home")
n_games <- nrow(home_away_only)

binom.test(home_wins, n_games, p = 0.5, alternative = "greater")
```

```
##
##  Exact binomial test
##
## data:  home_wins and n_games
## number of successes = 102, number of trials = 167, p-value = 0.002592
## alternative hypothesis: true probability of success is greater than 0.5
## 95 percent confidence interval:
##  0.5446038 1.0000000
## sample estimates:
## probability of success
##              0.6107784
```

When draws were excluded,home teams won 102 of the 167 matches that had a winner. This gives a home-win proportion of approximately 61.1%. The p-value was 0.002592, which is below 0.05, so there is statistically significant evidence that, among matches with a winner, home wins were more common than away wins.

Visualisation of match outcomes

```
ggplot(spfl, aes(x = Result)) +
  geom_bar() +
  labs(
    title = "Distribution of Match Outcomes",
    x = "Match Result",
    y = "Number of Matches"
  )
```

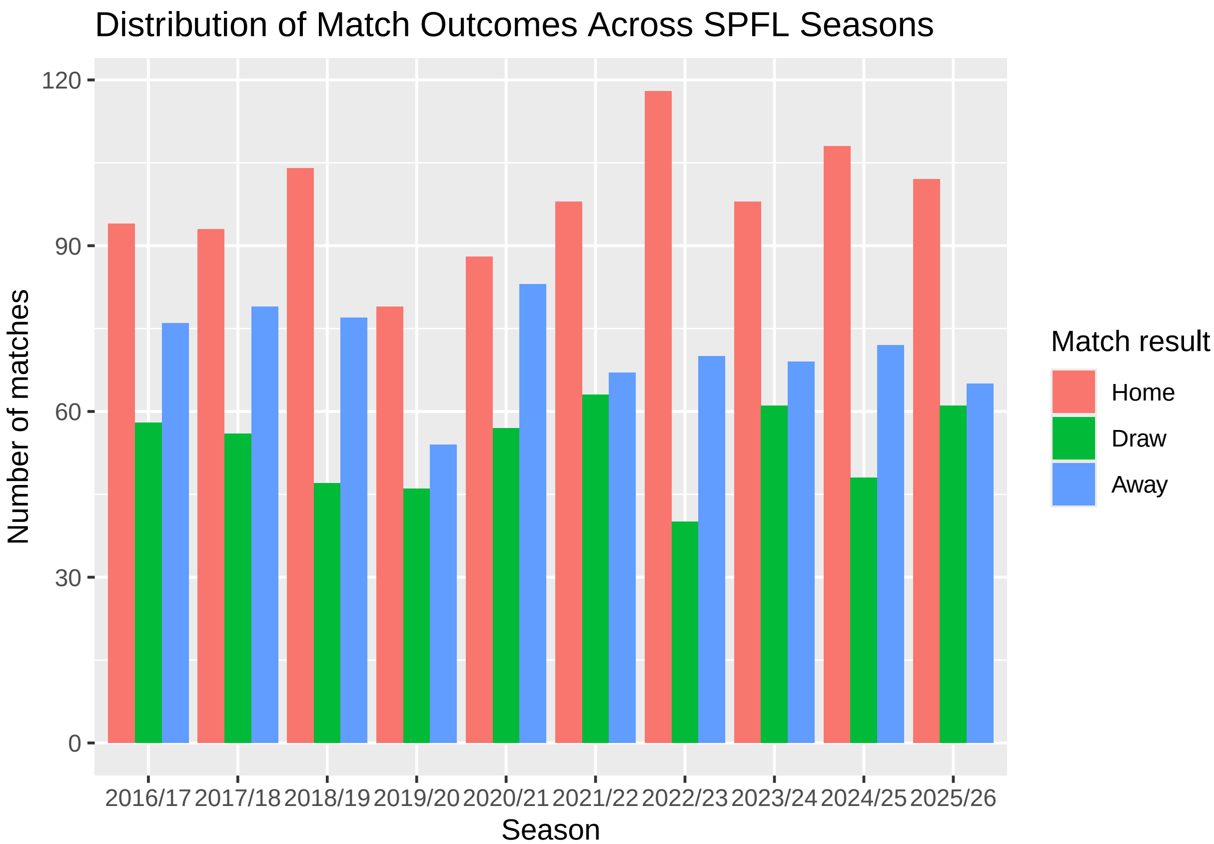



```r
# Combine the Season and Result columns from each individual season dataset.
# Only these two columns are needed for the match outcome distribution plot.
all_seasons_results <- bind_rows(
  spfl_2016_2017 %>% select(Season, Result),
  spfl_2017_2018 %>% select(Season, Result),
  spfl_2018_2019 %>% select(Season, Result),
  spfl_2019_2020 %>% select(Season, Result),
  spfl_2020_2021 %>% select(Season, Result),
  spfl_2021_2022 %>% select(Season, Result),
  spfl_2022_2023 %>% select(Season, Result),
  spfl_2023_2024 %>% select(Season, Result),
  spfl_2024_2025 %>% select(Season, Result),
  spfl_2025_2026 %>% select(Season, Result)
)

# Count the number of home wins, draws and away wins in each season.
match_outcomes_by_season <- all_seasons_results %>%

  # Keep only valid match result categories.
  filter(Result %in% c("Home", "Away", "Draw")) %>%

  # Set the order in which match results should appear in the plot.
  mutate(
    Result = factor(Result, levels = c("Home", "Draw", "Away"))
  ) %>%

  # Group by season and match outcome, then count the number of matches.
```

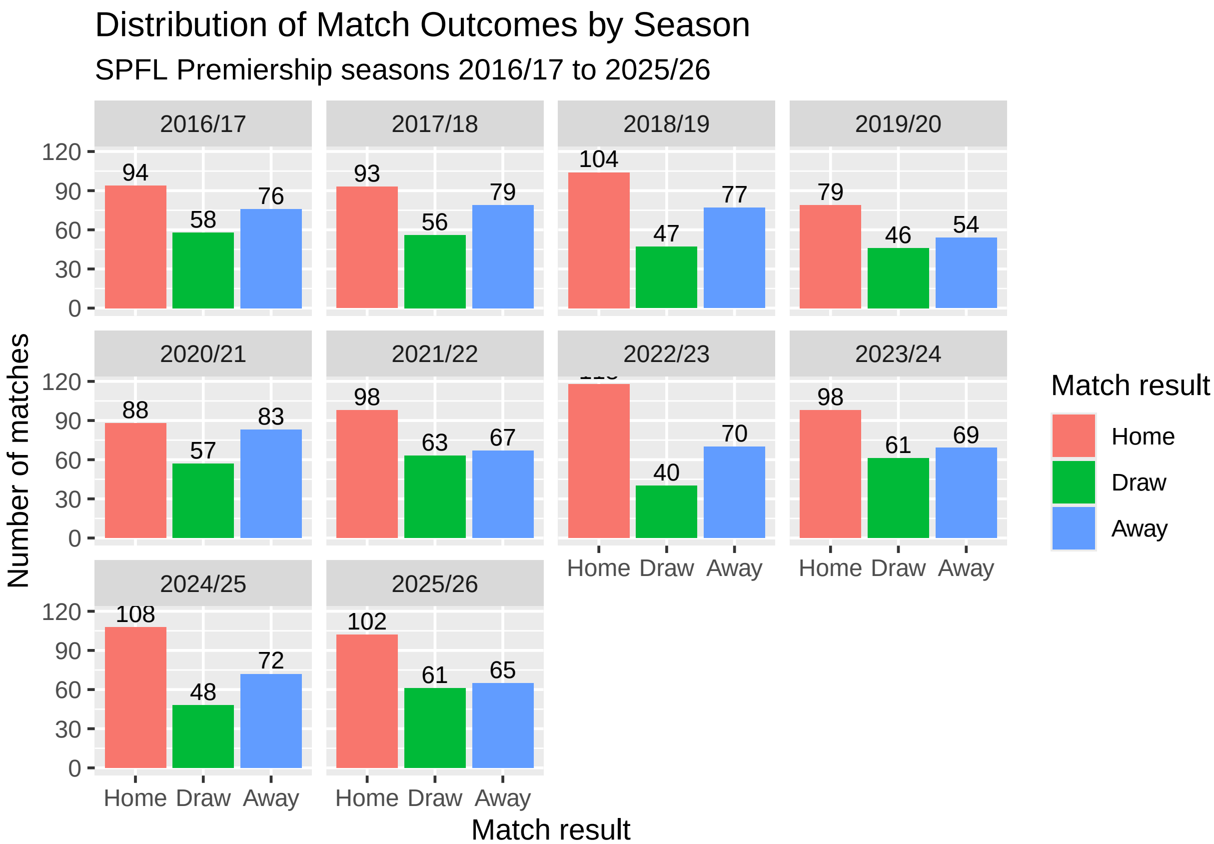
```r
  group_by(Season, Result) %>%
  summarise(
    NumberOfMatches = n(),
    .groups = "drop"
  )

match_outcomes_by_season
```

```
## # A tibble: 30 x 3
##    Season  Result NumberOfMatches
##    <chr>   <fct>            <int>
##  1 2016/17 Home                94
##  2 2016/17 Draw                58
##  3 2016/17 Away                76
##  4 2017/18 Home                93
##  5 2017/18 Draw                56
##  6 2017/18 Away                79
##  7 2018/19 Home               104
##  8 2018/19 Draw                47
##  9 2018/19 Away                77
## 10 2019/20 Home                79
## # i 20 more rows
```

```r
# Plot
ggplot(match_outcomes_by_season,
       aes(x = Season,
           y = NumberOfMatches,
           fill = Result)) +
  geom_col(position = "dodge") +
  labs(
    title = "Distribution of Match Outcomes Across SPFL Seasons",
    x = "Season",
    y = "Number of matches",
    fill = "Match result"
  )
```




```r
# Produce a faceted bar chart showing the distribution of match outcomes
# separately for each SPFL Premiership season.
ggplot(match_outcomes_by_season,
       aes(x = Result,
           y = NumberOfMatches,
           fill = Result)) +
  geom_col() +

  # Add the match count above each bar.
  geom_text(
    aes(label = NumberOfMatches),
    vjust = -0.3,
    size = 3
  ) +

  # Create a separate panel for each season.
  facet_wrap(~ Season) +
  labs(
    title = "Distribution of Match Outcomes by Season",
    subtitle = "SPFL Premiership seasons 2016/17 to 2025/26",
    x = "Match result",
    y = "Number of matches",
    fill = "Match result"
  )
```



Paired t-test: do home teams score more goals than away teams?

A paired t-test was used because each match contains a home-goal value and an away-goal value.

```r
t.test(spfl$`Home Goals`,
       spfl$`Away Goals`,
       paired = TRUE,
       alternative = "greater")
```

```
##
##  Paired t-test
##
## data:  spfl$`Home Goals` and spfl$`Away Goals`
## t = 3.5103, df = 227, p-value = 0.0002699
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
##  0.2090124       Inf
## sample estimates:
## mean difference
##       0.3947368
```

```r
mean(spfl$`Home Goals`, na.rm = TRUE)
```

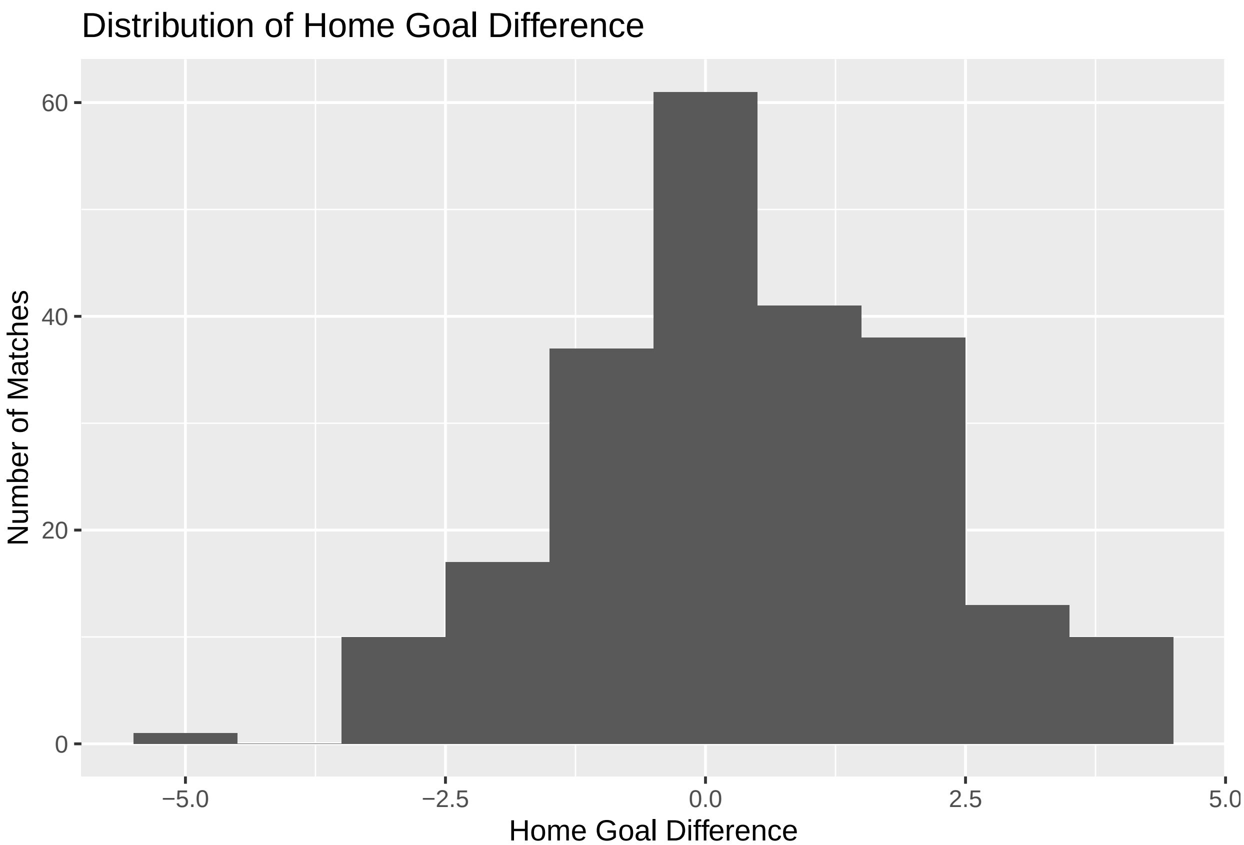
```
## [1] 1.587719
```

```r
mean(spfl$`Away Goals`, na.rm = TRUE)
```

```
## [1] 1.192982
```

The p-value was 0.0002699, which is less than 0.05. Therefore, there is sufficient evidence to suggest that home teams scored more goals than away teams.

```r
spfl$HomeGoalDifference <- spfl$`Home Goals` - spfl$`Away Goals`

ggplot(spfl, aes(x = HomeGoalDifference)) +
  geom_histogram(binwidth = 1) +
  labs(
    title = "Distribution of Home Goal Difference",
    x = "Home Goal Difference",
    y = "Number of Matches"
  )
```



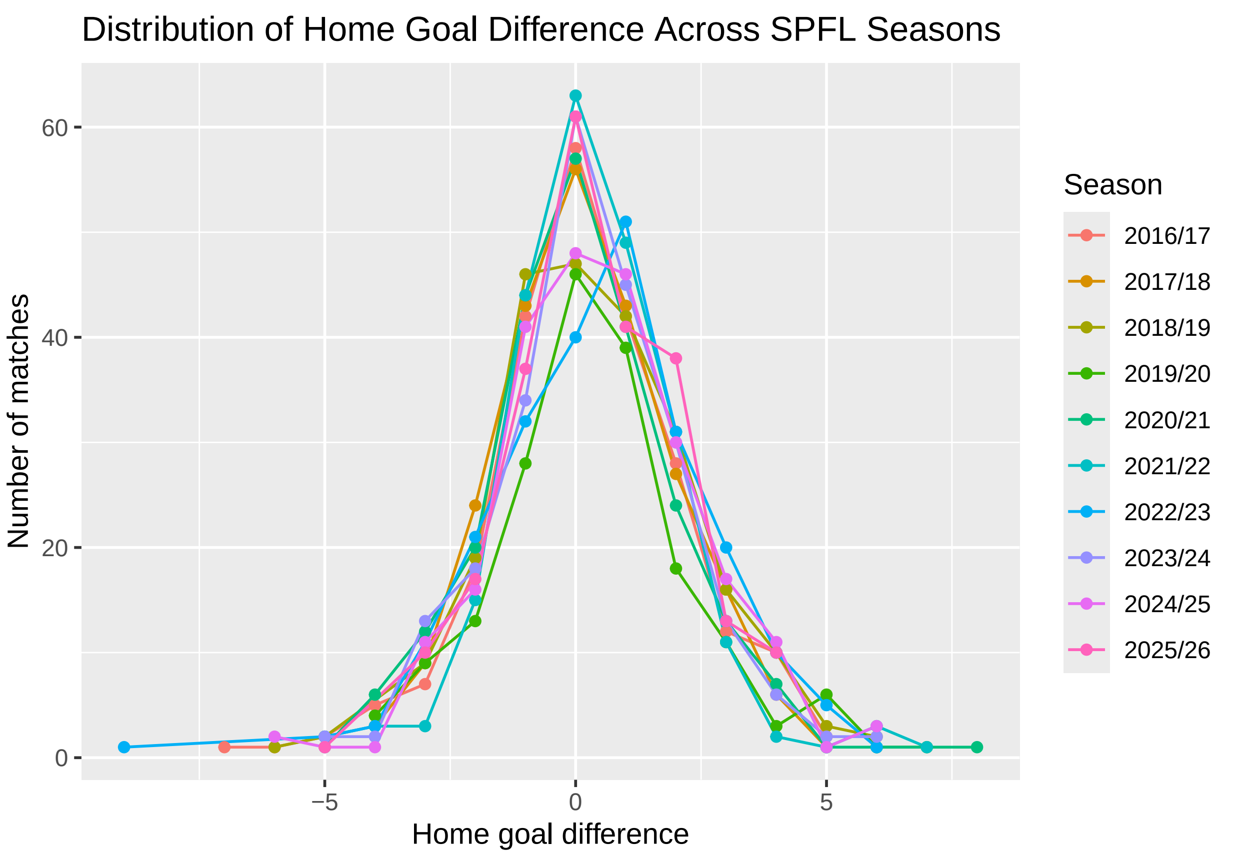

```r
library(readxl)
library(dplyr)

# Function to load one season and keep the columns needed
load_goal_difference_data <- function(file, season, sheet = NULL) {

  if (is.null(sheet)) {
    data <- read_excel(file)
  } else {
    data <- read_excel(file, sheet = sheet)
  }

  data %>%
    select(`Home Team`, `Away Team`, `Home Goals`, `Away Goals`, Result) %>%
    mutate(Season = season)
```

```r
}

# Load each season
spfl_2016_2017 <- load_goal_difference_data("SPFL 2016-2017.xlsx", "2016/17", "SPFL 2016-2017")
spfl_2017_2018 <- load_goal_difference_data("SPFL 2017-2018.xlsx", "2017/18", "SPFL 2017-2018")
spfl_2018_2019 <- load_goal_difference_data("SPFL 2018-2019.xlsx", "2018/19", "SPFL 2018-2019")
spfl_2019_2020 <- load_goal_difference_data("SPFL 2019-2020.xlsx", "2019/20", "SPFL 2019-2020")
spfl_2020_2021 <- load_goal_difference_data("SPFL 2020-2021.xlsx", "2020/21", "SPFL 2020-2021")
spfl_2021_2022 <- load_goal_difference_data("SPFL 2021-2022.xlsx", "2021/22")
spfl_2022_2023 <- load_goal_difference_data("SPFL 2022-2023.xlsx", "2022/23")
spfl_2023_2024 <- load_goal_difference_data("SPFL 2023-2024.xlsx", "2023/24")
spfl_2024_2025 <- load_goal_difference_data("SPFL 2024-2025.xlsx", "2024/25")
spfl_2025_2026 <- load_goal_difference_data("SPFL 2025-2026.xlsx", "2025/26", "SPFL Dataset")

# Combine all seasons
all_goal_difference_data <- bind_rows(
  spfl_2016_2017,
  spfl_2017_2018,
  spfl_2018_2019,
  spfl_2019_2020,
  spfl_2020_2021,
  spfl_2021_2022,
  spfl_2022_2023,
  spfl_2023_2024,
  spfl_2024_2025,
  spfl_2025_2026
)

# Calculate home goal difference
all_goal_difference_data <- all_goal_difference_data %>%
  mutate(
    HomeGoalDifference = `Home Goals` - `Away Goals`
  )

# Frequency table of home goal difference by season
home_goal_difference_all_seasons <- all_goal_difference_data %>%
  group_by(Season, HomeGoalDifference) %>%
  summarise(
    NumberOfMatches = n(),
    .groups = "drop"
  ) %>%
  arrange(Season, HomeGoalDifference)

home_goal_difference_all_seasons
```

```
## # A tibble: 120 x 3
##    Season  HomeGoalDifference NumberOfMatches
##    <chr>                <dbl>           <int>
##  1 2016/17                 -7               1
##  2 2016/17                 -6               1
##  3 2016/17                 -5               2
##  4 2016/17                 -4               5
##  5 2016/17                 -3               7
##  6 2016/17                 -2              18
```

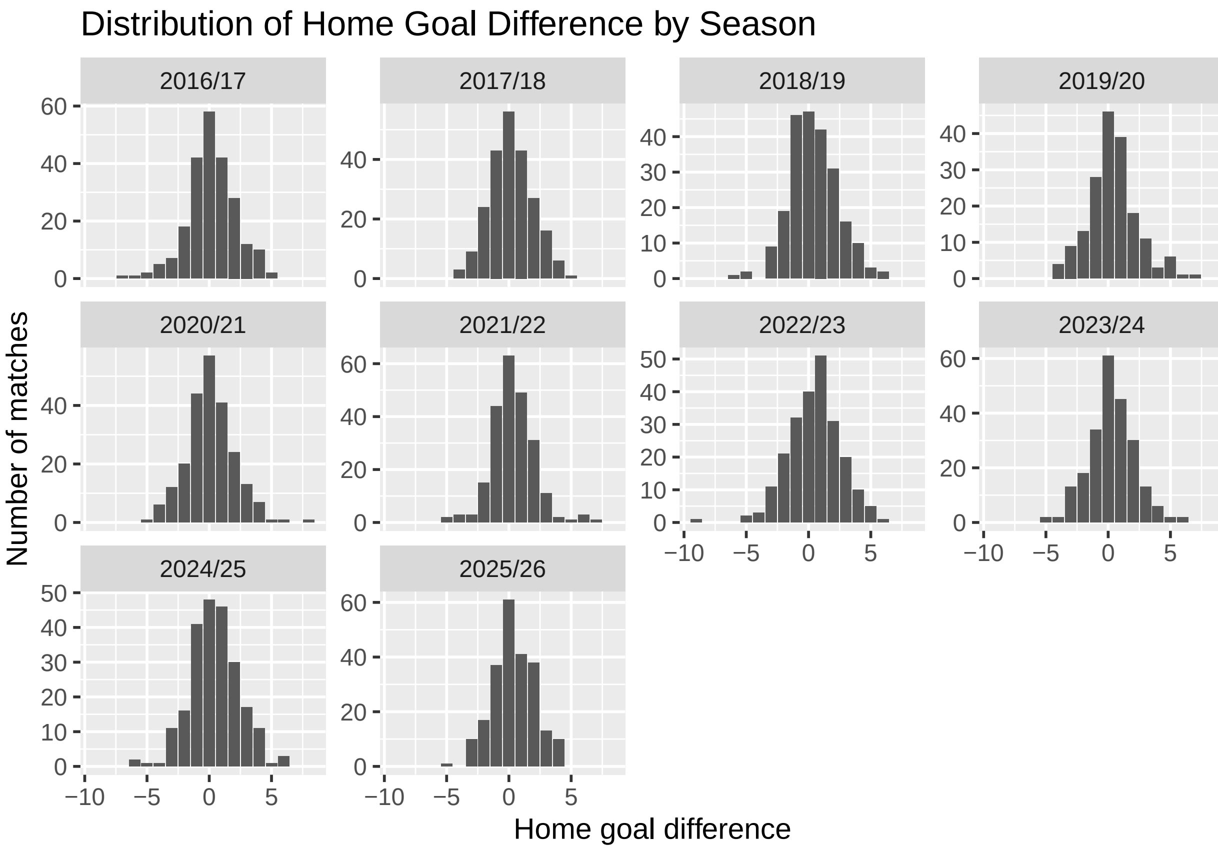
```
## 7 2016/17                   -1             42
## 8 2016/17                    0             58
## 9 2016/17                    1             42
## 10 2016/17                   2             28
## # i 110 more rows
```

```r
# Plot home goal difference distribution for all seasons
ggplot(home_goal_difference_all_seasons,
       aes(x = HomeGoalDifference,
           y = NumberOfMatches,
           colour = Season,
           group = Season)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Distribution of Home Goal Difference Across SPFL Seasons",
    x = "Home goal difference",
    y = "Number of matches",
    colour = "Season"
  )
```



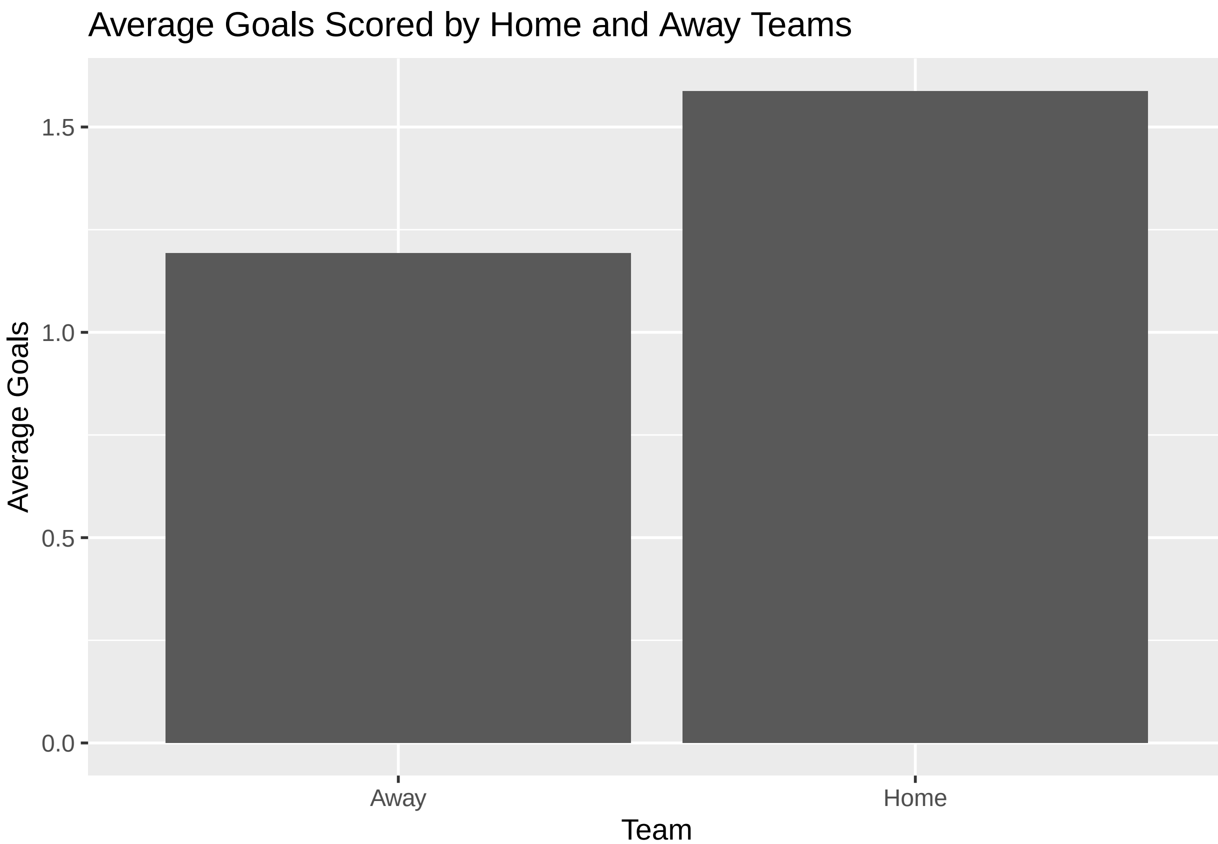

```r
  ggplot(home_goal_difference_all_seasons,
       aes(x = HomeGoalDifference, y = NumberOfMatches)) +
  geom_col() +
  facet_wrap(~ Season, scales = "free_y") +
  labs(
    title = "Distribution of Home Goal Difference by Season",
    x = "Home goal difference",
```

```
    y = "Number of matches"
  )
```

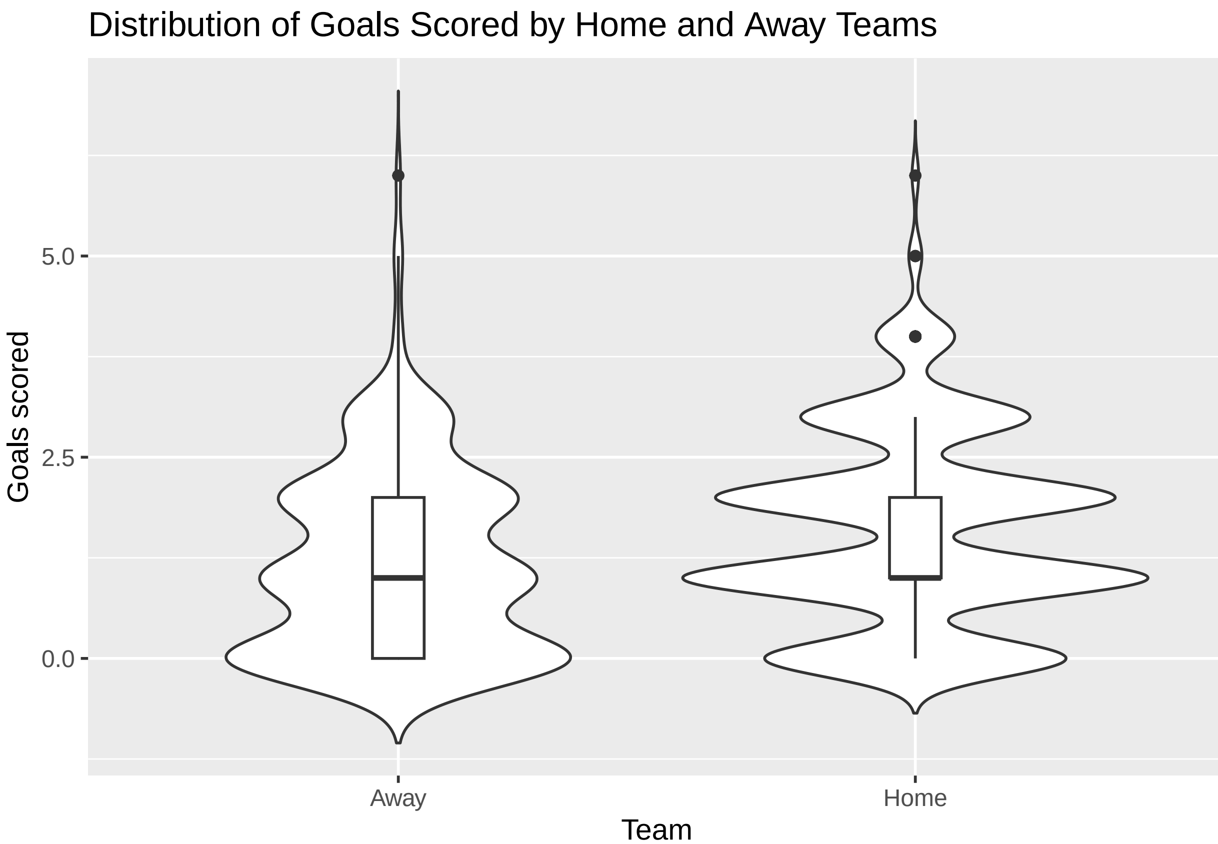


Visualisation of average home and away goals

```
# Calculate the average number of goals scored by home and away teams.
# na.rm = TRUE ensures that any missing goal values are ignored.
average_goals <- data.frame(
  Team = c("Home", "Away"),
  Goals = c(
    mean(spfl$`Home Goals`, na.rm = TRUE),
    mean(spfl$`Away Goals`, na.rm = TRUE)
  )
)

# Produce a bar chart comparing the average goals scored
# by home teams and away teams.
ggplot(average_goals, aes(x = Team, y = Goals)) +
  geom_col() +
  labs(
    title = "Average Goals Scored by Home and Away Teams",
    x = "Team",
    y = "Average Goals"
  )
```



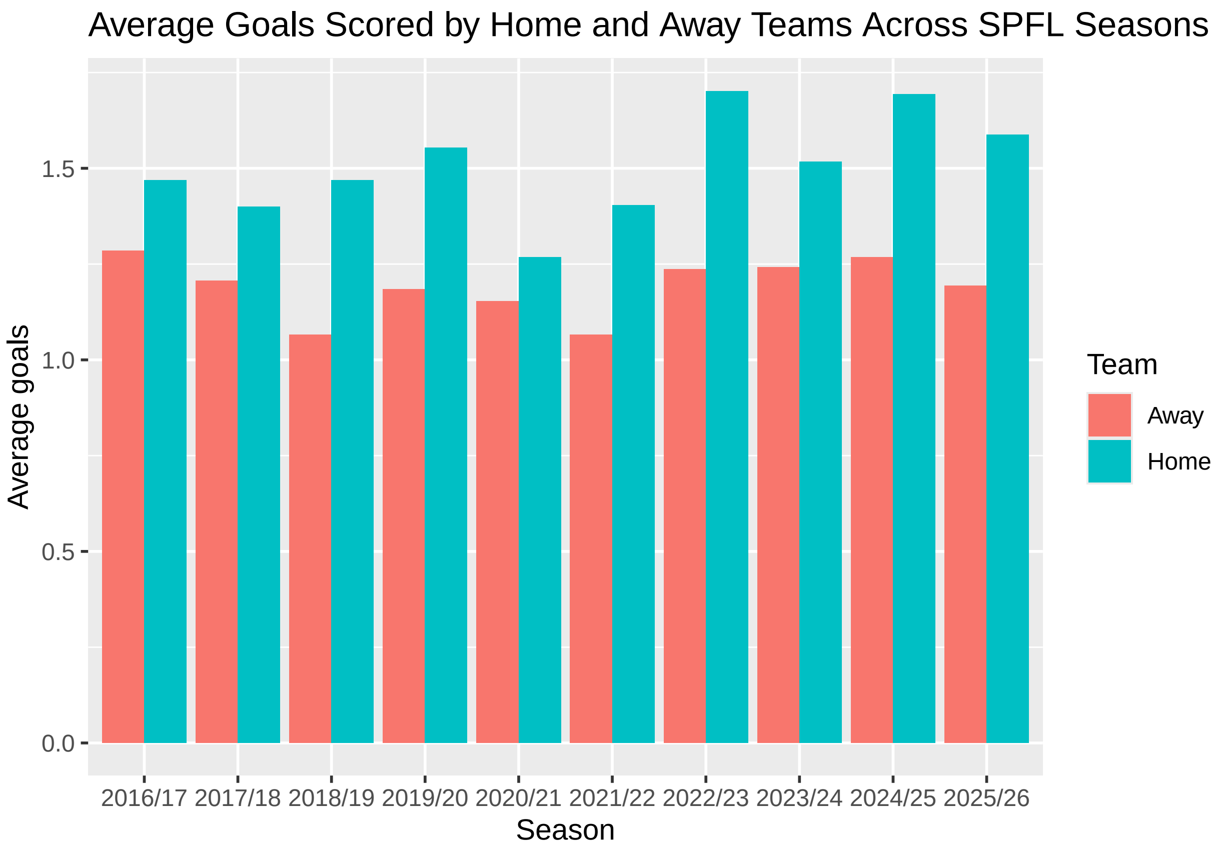

```r
goals_long <- data.frame(
  TeamType = c(rep("Home", nrow(spfl)), rep("Away", nrow(spfl))),
  Goals = c(spfl$`Home Goals`, spfl$`Away Goals`)
)

ggplot(goals_long,
       aes(x = TeamType, y = Goals)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.1) +
  labs(
    title = "Distribution of Goals Scored by Home and Away Teams",
    x = "Team",
    y = "Goals scored"
  )
```



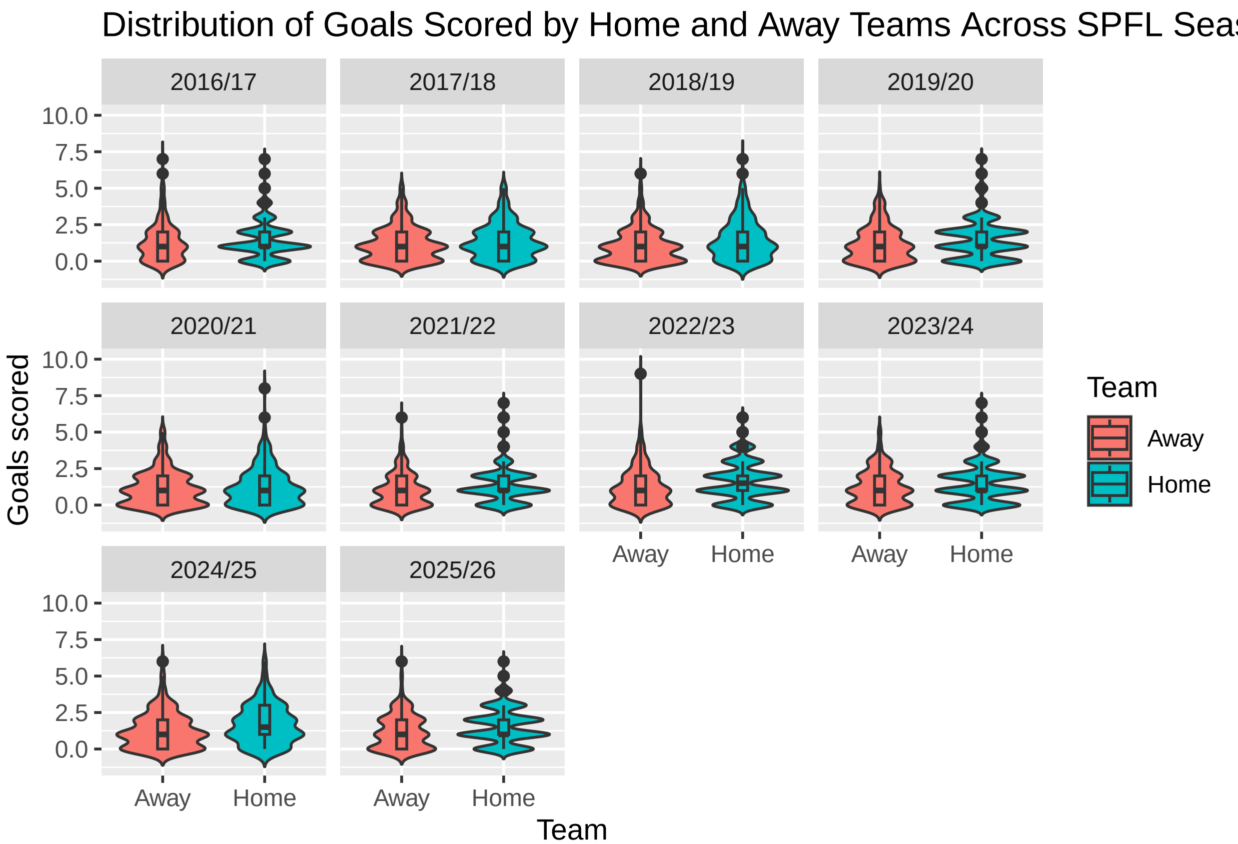

```r
# Create a long-format dataset with home and away goals in one column.
# This structure makes it easier to compare home and away goals across seasons.
average_goals_all_seasons <- data.frame(
  Season = c(
    all_goal_difference_data$Season,
    all_goal_difference_data$Season
  ),
  Team = c(
    rep("Home", nrow(all_goal_difference_data)),
    rep("Away", nrow(all_goal_difference_data))
  ),
  Goals = c(
    all_goal_difference_data$`Home Goals`,
    all_goal_difference_data$`Away Goals`
  )
)

# Calculate average goals by season and team type
average_goals_summary <- average_goals_all_seasons %>%
  group_by(Season, Team) %>%
  summarise(
    AverageGoals = mean(Goals, na.rm = TRUE),
    .groups = "drop"
  )

average_goals_summary
```

```
## # A tibble: 20 x 3
##    Season  Team  AverageGoals
##    <chr>   <chr>        <dbl>
##  1 2016/17 Away          1.29
##  2 2016/17 Home          1.47
##  3 2017/18 Away          1.21
##  4 2017/18 Home          1.40
##  5 2018/19 Away          1.07
##  6 2018/19 Home          1.47
##  7 2019/20 Away          1.18
##  8 2019/20 Home          1.55
##  9 2020/21 Away          1.15
## 10 2020/21 Home          1.27
## 11 2021/22 Away          1.07
## 12 2021/22 Home          1.40
## 13 2022/23 Away          1.24
## 14 2022/23 Home          1.70
## 15 2023/24 Away          1.24
## 16 2023/24 Home          1.52
## 17 2024/25 Away          1.27
## 18 2024/25 Home          1.69
## 19 2025/26 Away          1.19
## 20 2025/26 Home          1.59
```

```r
# Plot average home and away goals for every season
ggplot(average_goals_summary,
       aes(x = Season,
           y = AverageGoals,
           fill = Team)) +
  geom_col(position = "dodge") +
  labs(
    title = "Average Goals Scored by Home and Away Teams Across SPFL Seasons",
    x = "Season",
    y = "Average goals",
    fill = "Team"
  )
```

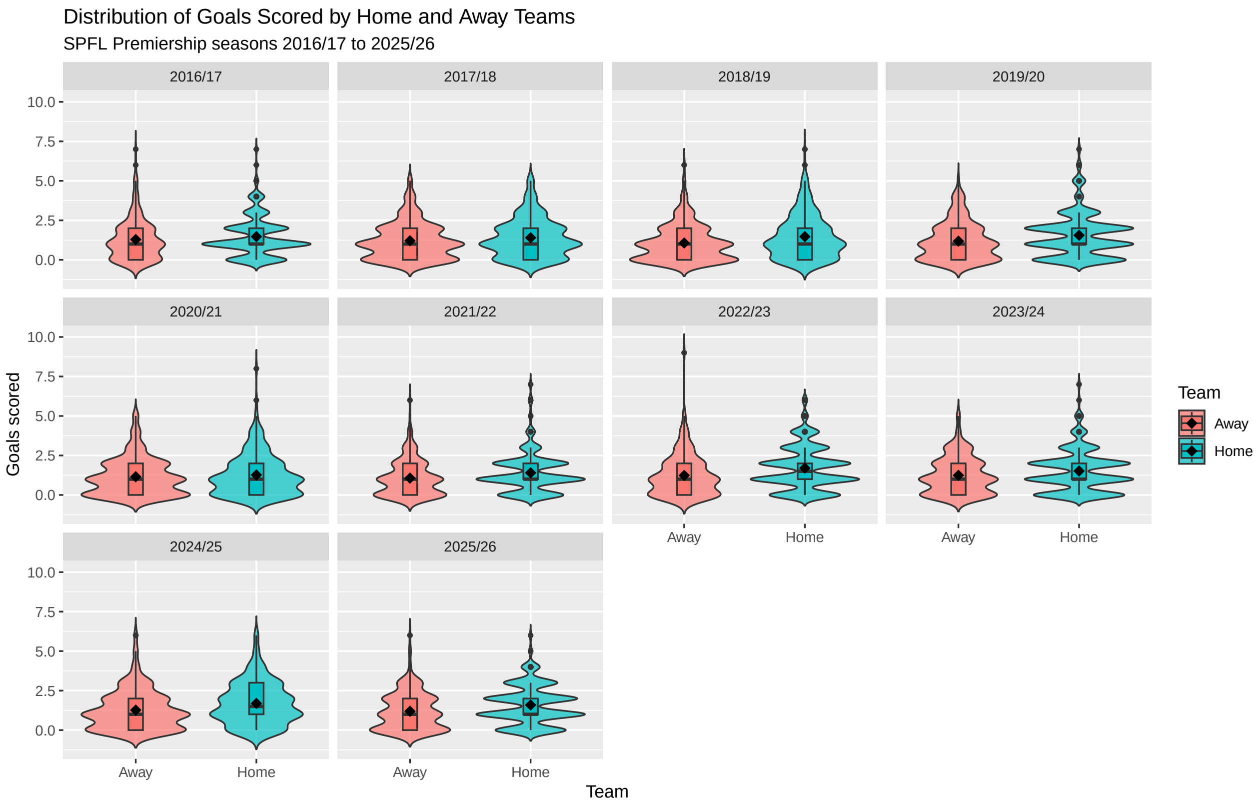



```r
ggplot(average_goals_all_seasons,
       aes(x = Team,
           y = Goals,
           fill = Team)) +
  geom_violin(trim = FALSE) +
  geom_boxplot(width = 0.1) +
  facet_wrap(~ Season) +
  labs(
    title = "Distribution of Goals Scored by Home and Away Teams Across SPFL Seasons",
    x = "Team",
    y = "Goals scored",
    fill = "Team"
  )
```



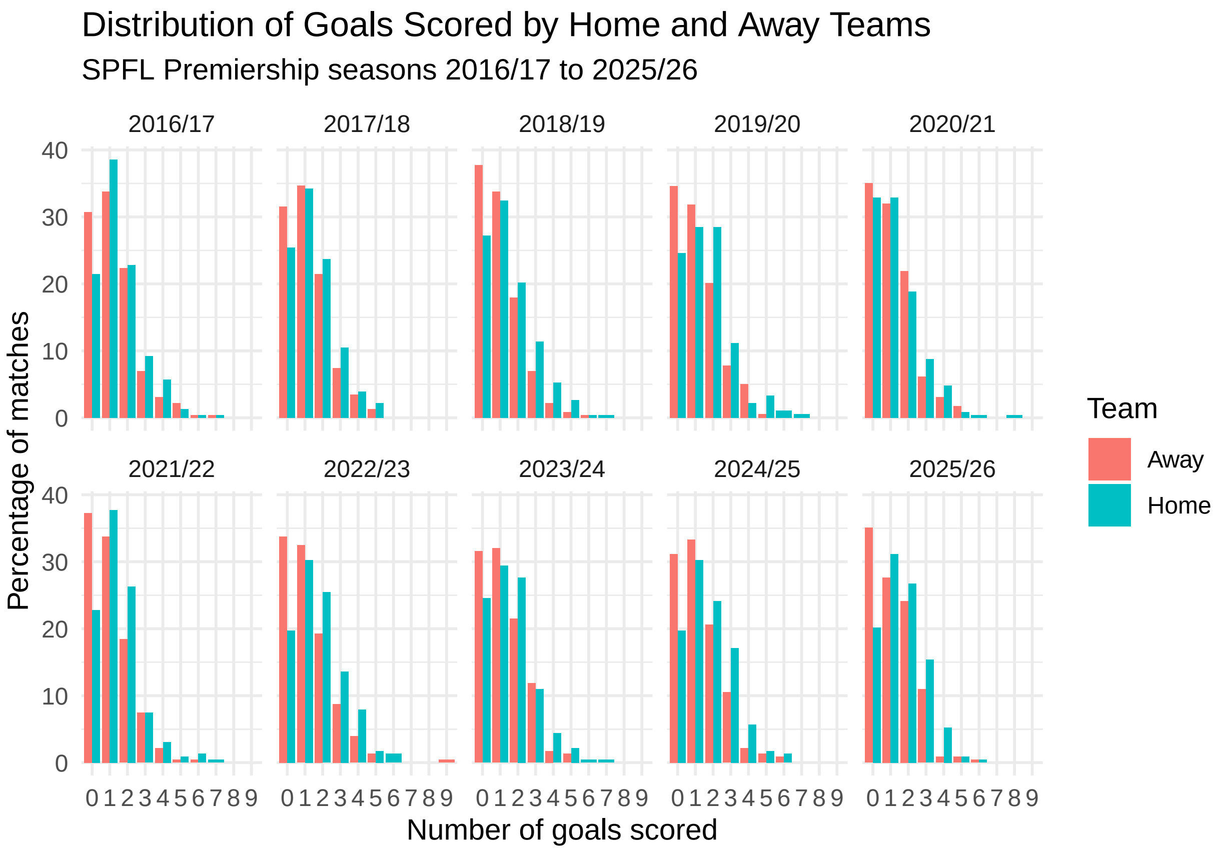

```r
# Produce a violin plot showing the distribution of goals scored
# by home and away teams across all seasons.
ggplot(average_goals_all_seasons,
       aes(x = Team,
           y = Goals,
           fill = Team)) +

  # Add violin plots to show the spread and density of goals scored.
  geom_violin(trim = FALSE, alpha = 0.7) +

  # Add boxplots inside the violins to show the median and interquartile range.
  geom_boxplot(width = 0.12, outlier.size = 1) +

  # Add a black diamond to show the mean number of goals scored.
  stat_summary(
    fun = mean,
    geom = "point",
    shape = 18,
    size = 3,
    colour = "black"
  ) +

  # Create a separate panel for each season.
  facet_wrap(~ Season, ncol = 4) +

  # Add plot labels.
  labs(
```

```
    title = "Distribution of Goals Scored by Home and Away Teams",
    subtitle = "SPFL Premiership seasons 2016/17 to 2025/26",
    x = "Team",
    y = "Goals scored",
    fill = "Team"
  )
```




```
# This code creates a bar plot showing the distribution of goals scored
# by home and away teams across each Scottish Premiership season.

# First, create a new dataset where home goals and away goals are placed
# into one single column called "Goals".
# This makes the data easier to plot using ggplot.
goals_distribution <- data.frame(

  # Repeat the season column twice: once for home goals and once for away goals
  Season = c(all_goal_difference_data$Season,
             all_goal_difference_data$Season),

  # Create a variable showing whether the goals were scored by the home or away team
  Team = c(rep("Home", nrow(all_goal_difference_data)),
           rep("Away", nrow(all_goal_difference_data))),

  # Combine the home goals and away goals into one column
  Goals = c(all_goal_difference_data$`Home Goals`,
            all_goal_difference_data$`Away Goals`)
)
```

```r
# Count how many times each number of goals was scored by home and away teams
# in each season.
goals_count <- goals_distribution %>%

  # Group the data by season, team type, and number of goals scored
  group_by(Season, Team, Goals) %>%

  # Count the number of matches in each group
  summarise(Count = n(), .groups = "drop") %>%

  # Group again by season and team so percentages are calculated separately
  # for home and away teams within each season
  group_by(Season, Team) %>%

  # Convert the raw counts into percentages of matches
  mutate(Percentage = Count / sum(Count) * 100)

# Create a bar plot showing the percentage distribution of goals scored
# by home and away teams for each season.
ggplot(goals_count, aes(x = factor(Goals, levels = 0:9),
                        y = Percentage,
                        fill = Team)) +

  # Create bars using the percentage values already calculated above.
  # position = "dodge" places the home and away bars side by side.
  geom_bar(stat = "identity", position = "dodge") +

  # Split the plot into separate panels for each season.
  # ncol = 5 gives two rows of five seasons.
  facet_wrap(~ Season, ncol = 5) +

  # Keep the x-axis goal values from 0 to 9 visible across all panels,
  # even if some goal totals do not occur in a particular season.
  scale_x_discrete(drop = FALSE) +

  # Add the title, subtitle, axis labels, and legend title.
  labs(
    title = "Distribution of Goals Scored by Home and Away Teams",
    subtitle = "SPFL Premiership seasons 2016/17 to 2025/26",
    x = "Number of goals scored",
    y = "Percentage of matches",
    fill = "Team"
  ) +

  # Use a simple, clean theme for the plot.
  theme_minimal()
```



Attendance and match outcome

Attendance in home wins compared with away wins

```r
home_win_attendance <- spfl$Attendance[spfl$Result == "Home"]
away_win_attendance <- spfl$Attendance[spfl$Result == "Away"]

t.test(home_win_attendance, away_win_attendance)
```

```
##
##  Welch Two Sample t-test
##
## data:  home_win_attendance and away_win_attendance
## t = 3.4345, df = 162.99, p-value = 0.000753
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##   3818.266 14147.578
## sample estimates:
## mean of x mean of y
##  22724.46  13741.54
```

The p-value was 0.000753. Matches won by the home team had significantly higher average attendances than matches won by the away team. However, this does not prove that attendance caused the home win, since larger clubs may both attract higher crowds and win more matches.

The mean attendances were:

```r
mean(home_win_attendance, na.rm = TRUE)
```

```
## [1] 22724.46
```

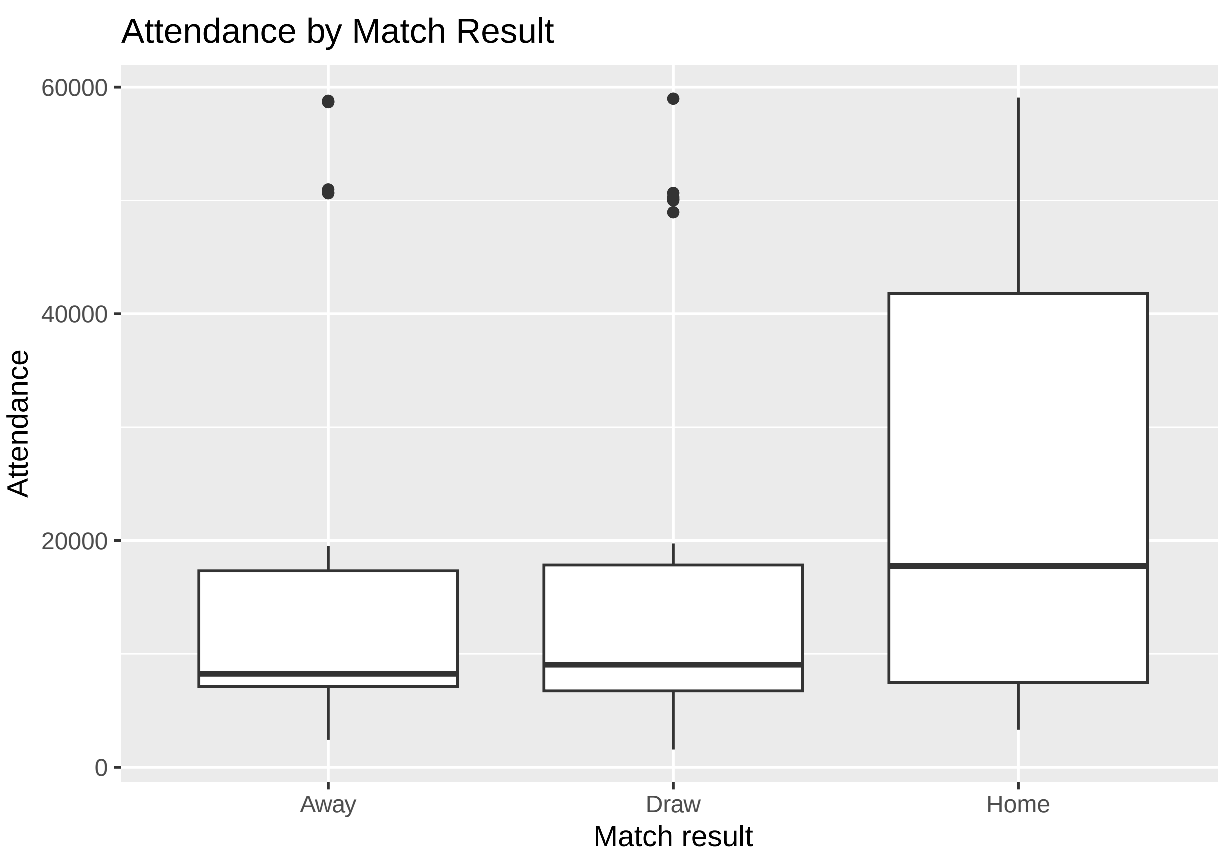
```r
mean(away_win_attendance, na.rm = TRUE)
```

```
## [1] 13741.54
```

The mean home-win attendance was approximately 22,724.46, while the mean away-win attendance was approximately 13,741.54.

Visualisation of attendance by match result

```r
# Produce a boxplot showing how attendance varies across match outcomes.
# This compares the distribution of attendances for home wins, draws and away wins.
ggplot(spfl, aes(x = Result, y = Attendance)) +

  # Add boxplots to show the median, interquartile range and possible outliers
  # in attendance for each match result category.
  geom_boxplot() +

  # Add plot labels.
  labs(
    title = "Attendance by Match Result",
    x = "Match result",
    y = "Attendance"
  )
```



Percent capacity in home wins compared with away wins

```r
home_win_capacity <- spfl$`Percent Capacity`[spfl$Result == "Home"]
away_win_capacity <- spfl$`Percent Capacity`[spfl$Result == "Away"]

t.test(home_win_capacity, away_win_capacity)
```

```
##
##  Welch Two Sample t-test
##
## data:  home_win_capacity and away_win_capacity
## t = 1.2736, df = 130.27, p-value = 0.2051
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.02470251  0.11398274
## sample estimates:
## mean of x mean of y
## 0.8029243 0.7582841
```

The p-value was 0.2051, which is greater than 0.05. Therefore, there is insufficient evidence to say that percent capacity differs significantly between home wins and away wins. Unlike raw attendance, percent capacity was not significantly different between home wins and away wins. This suggests that the raw attendance result may partly reflect differences in club size and stadium capacity rather than crowd intensity alone.

The mean percent capacities were:

```r
mean(home_win_capacity, na.rm = TRUE)
```

```
## [1] 0.8029243
```

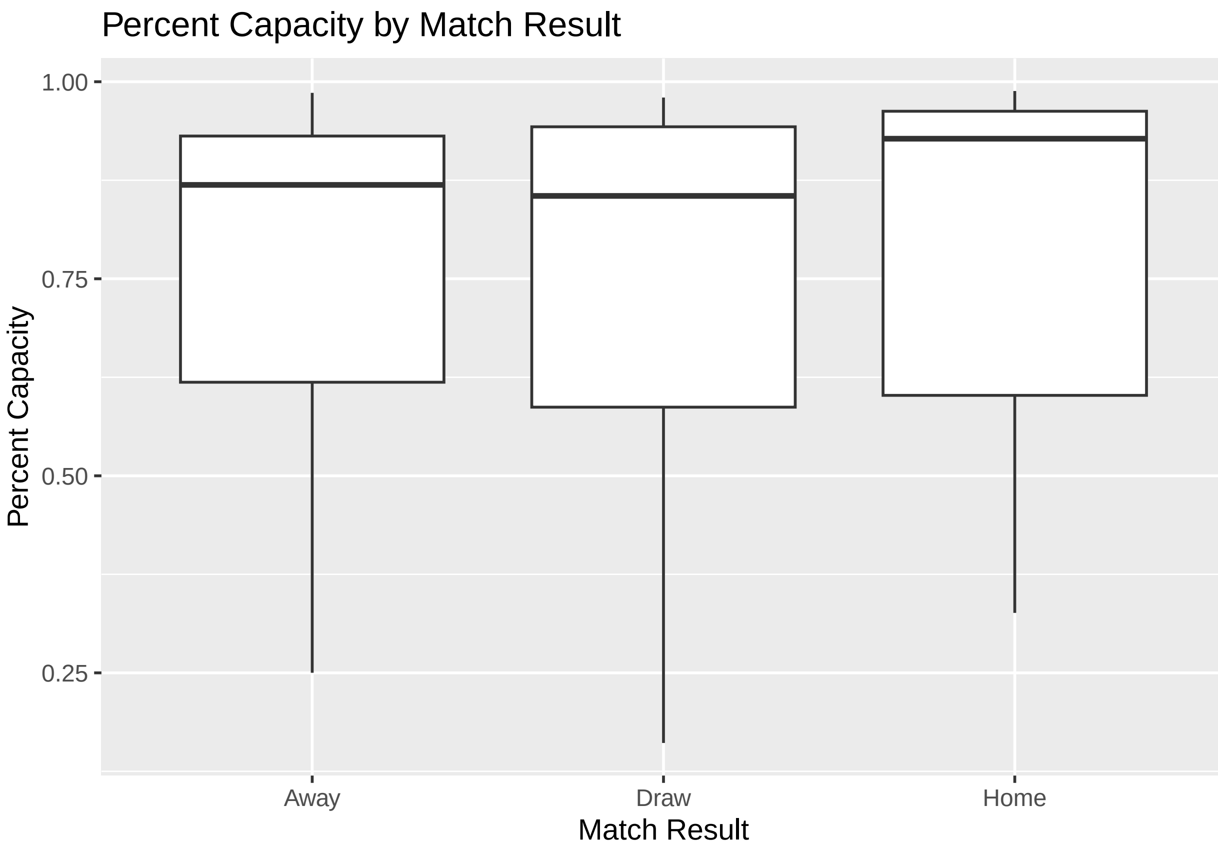
```r
mean(away_win_capacity, na.rm = TRUE)
```

```
## [1] 0.7582841
```

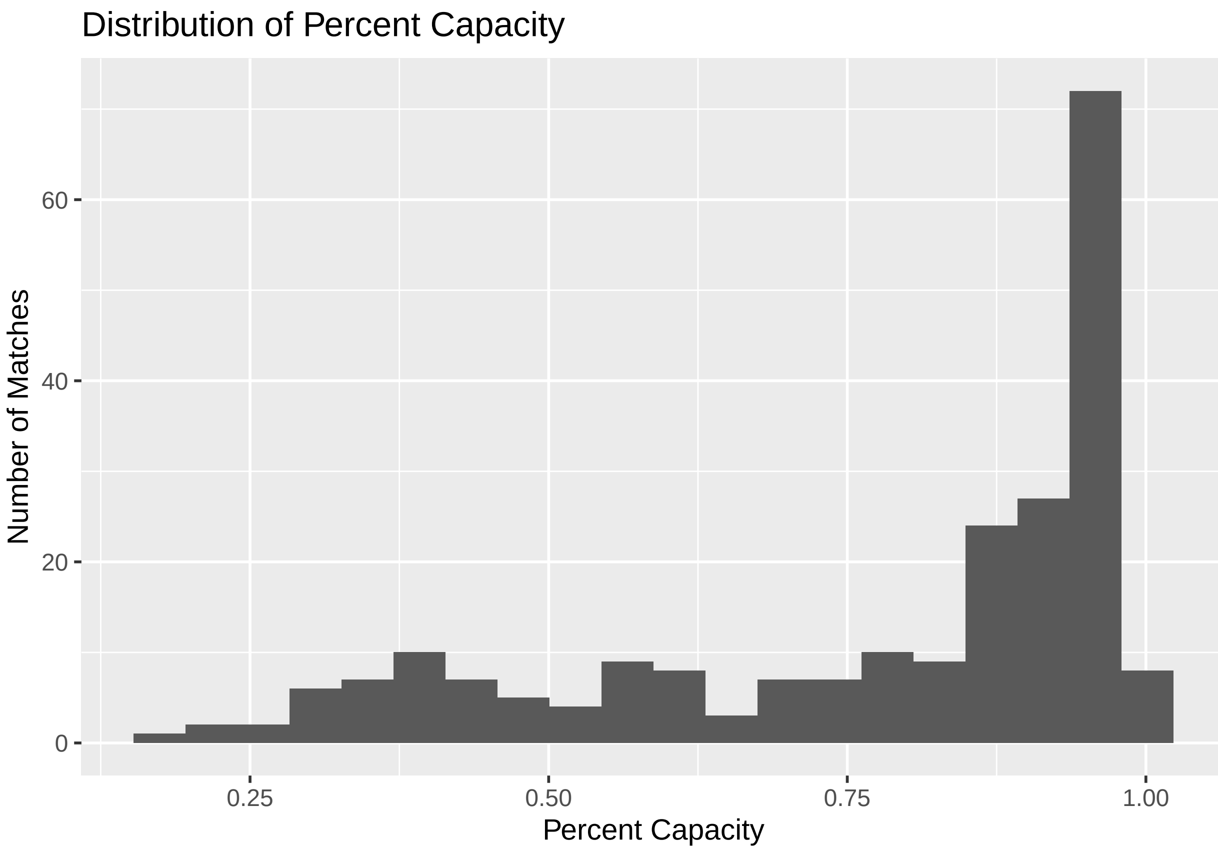
The mean home-win capacity was approximately 0.8029, while the mean away-win capacity was approximately 0.7583.

Visualisation of percent capacity by match result

```r
ggplot(spfl, aes(x = Result, y = `Percent Capacity`)) +
  geom_boxplot() +
  labs(
    title = "Percent Capacity by Match Result",
    x = "Match Result",
    y = "Percent Capacity"
  )
```




```r
ggplot(spfl, aes(x = `Percent Capacity`)) +
  geom_histogram(bins = 20) +
  labs(
    title = "Distribution of Percent Capacity",
    x = "Percent Capacity",
    y = "Number of Matches"
  )
```



Logistic regression: do percent capacity and travel distance affect the probability of a home win?

A logistic regression model was fitted because `HomeWin` is a binary response variable. The generalised linear model keeps predicted probabilities between 0 and 1.

```
# Create a binary outcome variable for whether the home team won.
# Home wins are coded as 1, while away wins and draws are coded as 0.
# Any missing or unexpected result values are coded as NA.
spfl$HomeWin <- ifelse(spfl$Result == "Home", 1,
                       ifelse(spfl$Result %in% c("Away", "Draw"), 0, NA))
```

```
# Fit a logistic regression model to examine whether percentage capacity
# and away-team travel distance are associated with the probability
# of a home win.
logit_model <- glm(
  HomeWin ~ `Percent Capacity` + `Travel Distance for Away Team`,
  data = spfl,
  family = "binomial"
)

# Display the model output, including coefficient estimates,
# standard errors, z-values and p-values.
summary(logit_model)
```

```
##
## Call:
```

```
## glm(formula = HomeWin ~ `Percent Capacity` + `Travel Distance for Away Team`,
##     family = "binomial", data = spfl)
##
## Coefficients:
##                                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)                     -1.521117   0.592337  -2.568   0.0102 *
## `Percent Capacity`               1.247807   0.636133   1.962   0.0498 *
## `Travel Distance for Away Team`  0.005562   0.003317   1.677   0.0936 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 313.54  on 227  degrees of freedom
## Residual deviance: 307.78  on 225  degrees of freedom
## AIC: 313.78
##
## Number of Fisher Scoring iterations: 4
```

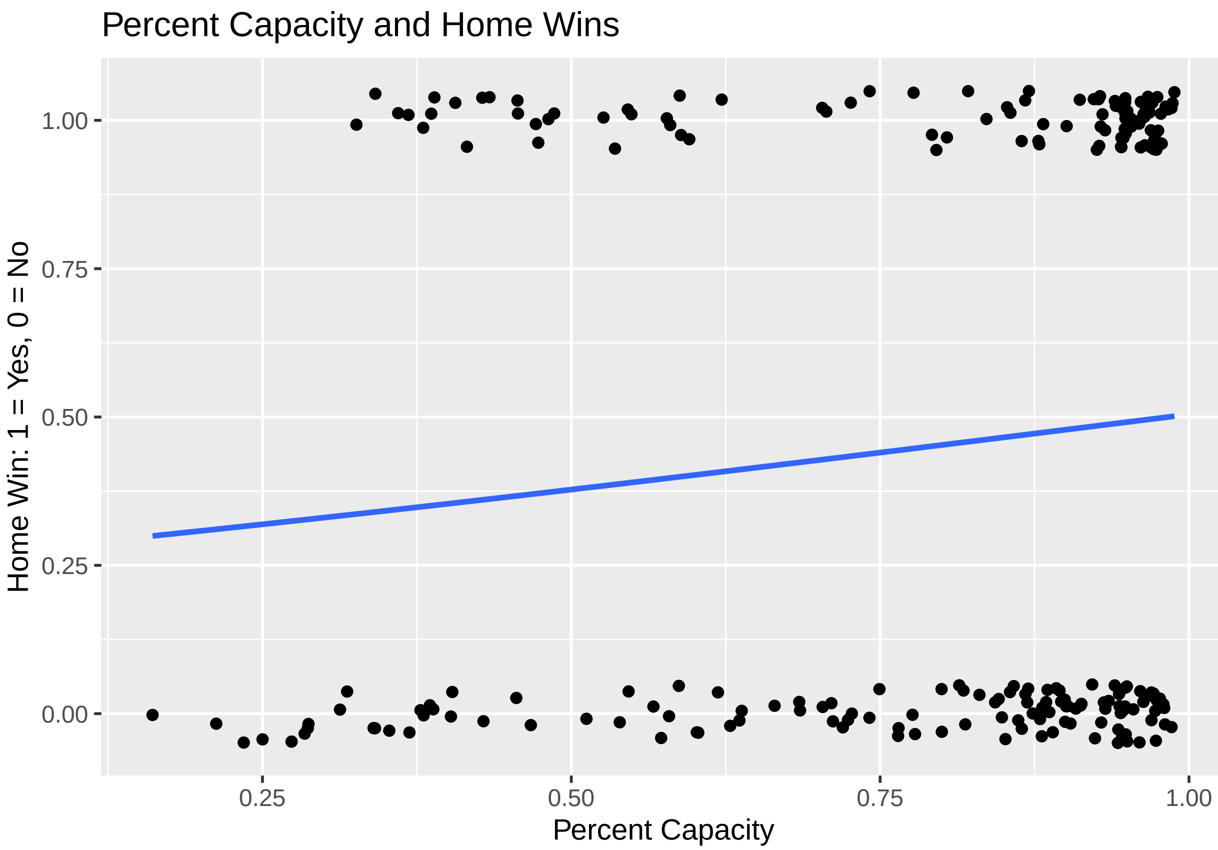
```r
# Show one-sided p-values
pnorm(
  summary(logit_model)$coefficients[, "z value"],
  lower.tail = FALSE
)
```

```
##                     (Intercept)              `Percent Capacity`
##                      0.99488553                      0.02490746
## `Travel Distance for Away Team`
##                      0.04681547
```

One-sided hypothesis tests were used because the analysis tested the directional hypotheses that greater percentage capacity and greater away-team travel distance would increase the probability of a home win.

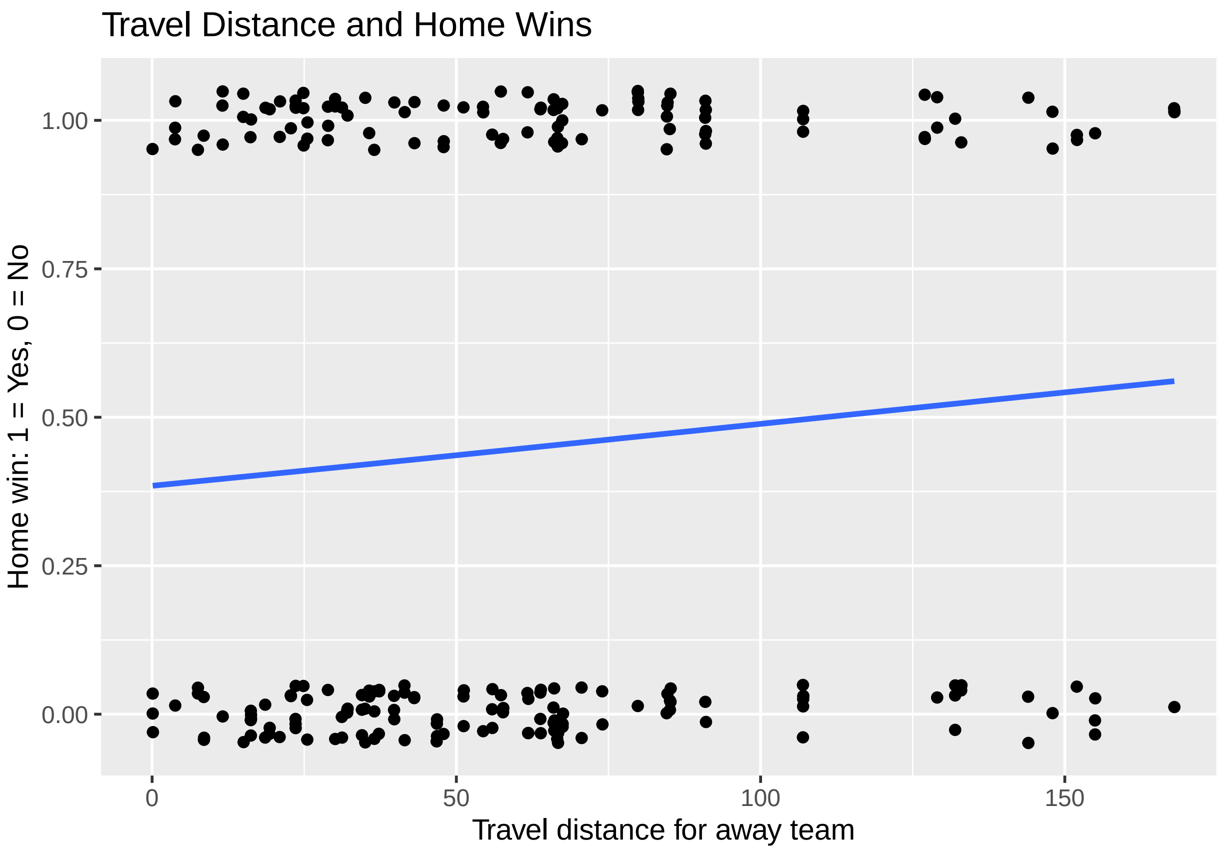
Visualisation of percent capacity and home wins

```r
ggplot(spfl, aes(x = `Percent Capacity`, y = HomeWin)) +
  geom_jitter(height = 0.05) +
  geom_smooth(method = "glm", method.args = list(family = "binomial"), se = FALSE) +
  labs(
    title = "Percent Capacity and Home Wins",
    x = "Percent Capacity",
    y = "Home Win: 1 = Yes, 0 = No"
  )
```



Visualisation of travel distance and home wins

```
# Produce a plot showing the relationship between away-team travel distance
# and whether the home team won the match.
ggplot(spfl, aes(x = `Travel Distance for Away Team`, y = HomeWin)) +

  # Add jittered points so that overlapping 0 and 1 values are easier to see.
  # HomeWin is binary, where 1 indicates a home win and 0 indicates no home win.
  geom_jitter(height = 0.05) +

  # Add a logistic regression trend line because the response variable,
  # HomeWin, is binary.
  geom_smooth(
    method = "glm",
    method.args = list(family = "binomial"),
    se = FALSE
  ) +

  # Add plot labels.
  labs(
    title = "Travel Distance and Home Wins",
    x = "Travel distance for away team",
    y = "Home win: 1 = Yes, 0 = No"
  )
```



Disciplinary analysis: did away teams receive more cards?

Yellow cards

```
t.test(spfl$`Away Yellow cards`,
       spfl$`Home Yellow cards`,
       paired = TRUE,
       alternative = "greater")
```

```
##
##  Paired t-test
##
## data:  spfl$`Away Yellow cards` and spfl$`Home Yellow cards`
## t = 2.9709, df = 227, p-value = 0.001644
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
##  0.1558162       Inf
## sample estimates:
## mean difference
##       0.3508772
```

The p-value was 0.001644, which is below 0.05. Therefore, the null hypothesis is rejected. There is statistically significant evidence that away teams received more yellow cards than home teams. Since cards are count data, a Poisson or negative binomial model could be used in a more advanced analysis. However, the paired t-test provides a simple preliminary comparison of home and away disciplinary outcomes within the same matches.

Distribution of yellow cards per match

```r
home_yellows <- table(spfl$`Home Yellow cards`)
away_yellows <- table(spfl$`Away Yellow cards`)

home_yellows
```

```
##
##  0  1  2  3  4  5  6  7
## 38 60 69 39 17  2  2  1
```

```r
away_yellows
```

```
##
##  0  1  2  3  4  5  6
## 23 55 68 45 23 10  4
```

```r
home_yellows_covid <- table(spfl_2020_2021$`Home Yellow cards`)
away_yellows_covid <- table(spfl_2020_2021$`Away Yellow cards`)
```

```r
yellow_values <- 0:max(
  spfl$`Home Yellow cards`,
  spfl$`Away Yellow cards`,
  na.rm = TRUE
)

yellow_card_table <- data.frame(
  YellowCards = yellow_values,
  Home = as.numeric(table(factor(spfl$`Home Yellow cards`,
                               levels = yellow_values))),
  Away = as.numeric(table(factor(spfl$`Away Yellow cards`,
                               levels = yellow_values)))
)

yellow_card_table
```

```
##   YellowCards Home Away
## 1           0   38   23
## 2           1   60   55
## 3           2   69   68
## 4           3   39   45
## 5           4   17   23
## 6           5    2   10
## 7           6    2    4
## 8           7    1    0
```

```r
# Reload full 2020/21 dataset because yellow-card columns are needed
spfl_2020_full <- read_excel("SPFL 2020-2021.xlsx", sheet = "SPFL 2020-2021")

# Check exact column names if needed
names(spfl_2020_full)
```

```
##  [1] "Date"               "Home Team"          "Away Team"
##  [4] "Home Goals"         "Away Goals"         "Result"
##  [7] "Goal Difference"    "Home Yellow cards" "Home Red cards"
## [10] "Away Yellow cards" "Away Red cards"
```

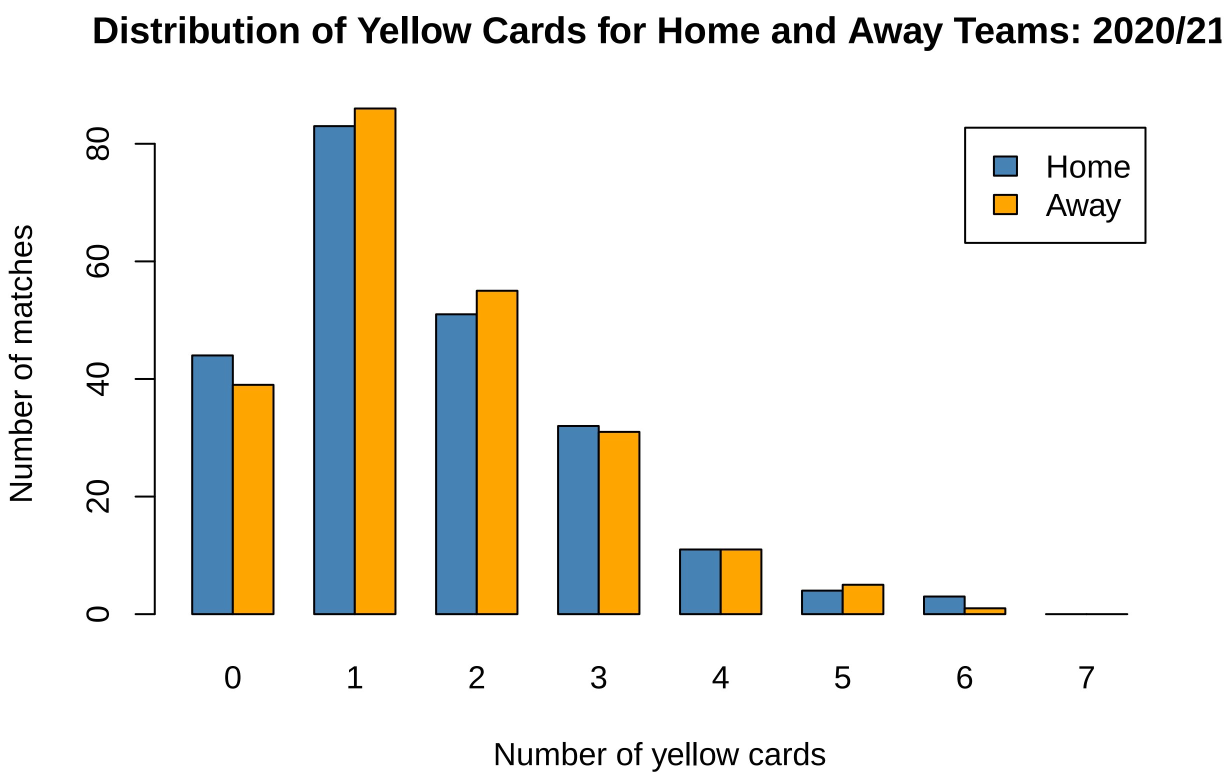
```r
# Create yellow card values
yellow_values_covid <- 0:max(
  c(
    spfl_2020_full$`Home Yellow cards`,
    spfl_2020_full$`Away Yellow cards`
  ),
  na.rm = TRUE
)

# Combined yellow-card table
yellow_card_table_covid <- data.frame(
  YellowCards = yellow_values_covid,
  Home = as.numeric(table(factor(
    spfl_2020_full$`Home Yellow cards`,
    levels = yellow_values_covid
  ))),
  Away = as.numeric(table(factor(
    spfl_2020_full$`Away Yellow cards`,
    levels = yellow_values_covid
  )))
)

yellow_card_table_covid
```

```
##   YellowCards Home Away
## 1           0   44   39
## 2           1   83   86
## 3           2   51   55
## 4           3   32   31
## 5           4   11   11
## 6           5    4    5
## 7           6    3    1
```

Visualisation of yellow cards

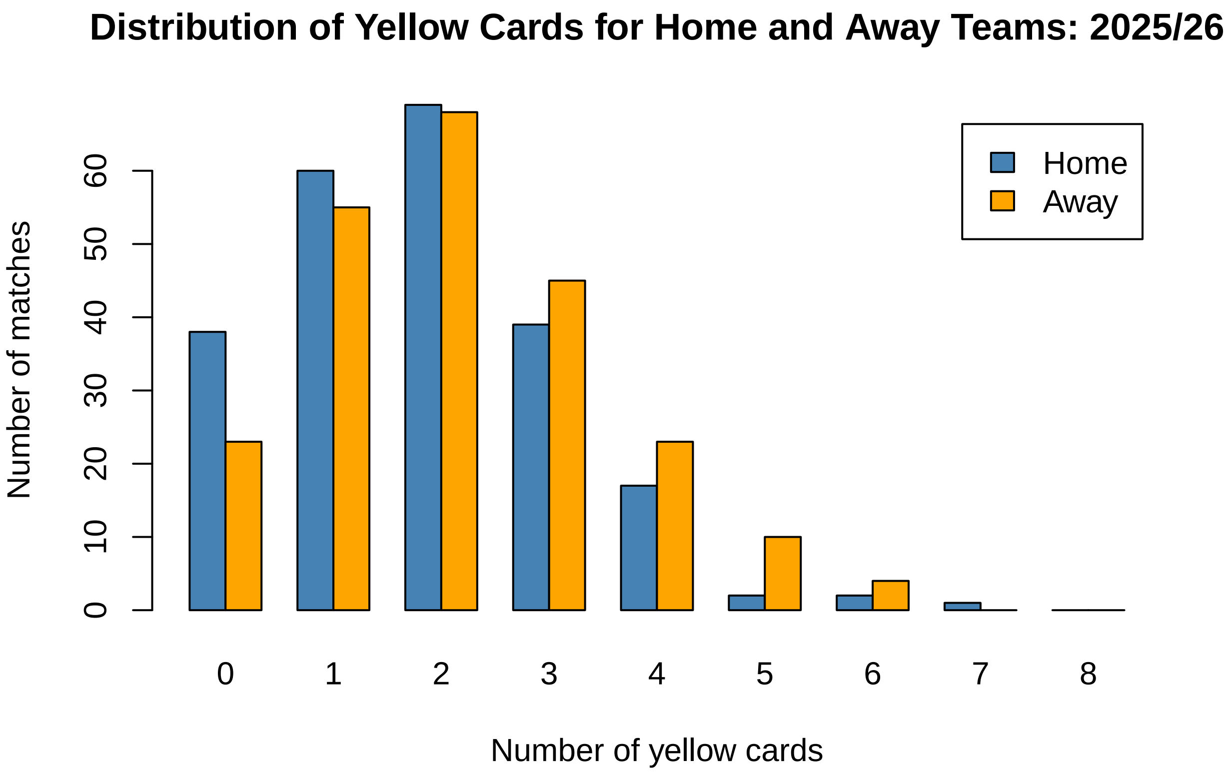
```r
# Reload the full 2020/21 dataset because the yellow-card columns
# are required for this analysis.
spfl_2020_full <- read_excel("SPFL 2020-2021.xlsx", sheet = "SPFL 2020-2021")

# Define the possible yellow-card values to be displayed on the plot.
# This ensures that all values from 0 to 7 appear, even if some have zero counts.
# Most yellow cards in a single game was 7
yellow_values_covid <- 0:7

# Count how many matches had each number of home-team yellow cards.
home_yellows_covid <- table(
  factor(spfl_2020_full$`Home Yellow cards`, levels = yellow_values_covid)
)

# Count how many matches had each number of away-team yellow cards.
away_yellows_covid <- table(
  factor(spfl_2020_full$`Away Yellow cards`, levels = yellow_values_covid)
)
```

```r
# Produce a grouped bar chart comparing the distribution of yellow cards
# for home and away teams during the 2020/21 season.
barplot(
  rbind(home_yellows_covid, away_yellows_covid),
  beside = TRUE,
  legend.text = c("Home", "Away"),
  col = c("steelblue", "orange"),
  main = "Distribution of Yellow Cards for Home and Away Teams: 2020/21",
  xlab = "Number of yellow cards",
  ylab = "Number of matches"
)
```




```r
# Define the possible yellow-card values to be displayed on the plot.
# This keeps the x-axis consistent, including values with zero matches.
yellow_values <- 0:8

# Count how many 2025/26 matches had each number of home-team yellow cards.
home_yellows <- table(
  factor(spfl$`Home Yellow cards`, levels = yellow_values)
)

# Count how many 2025/26 matches had each number of away-team yellow cards.
away_yellows <- table(
  factor(spfl$`Away Yellow cards`, levels = yellow_values)
)

# Produce a grouped bar chart comparing the distribution of yellow cards
# for home and away teams during the 2025/26 season.
barplot(
```

```
  rbind(home_yellows, away_yellows),
  beside = TRUE,
  legend.text = c("Home", "Away"),
  col = c("steelblue", "orange"),
  main = "Distribution of Yellow Cards for Home and Away Teams: 2025/26",
  xlab = "Number of yellow cards",
  ylab = "Number of matches"
)
```




```
yellow_values <- 0:max(
  spfl$`Home Yellow cards`,
  spfl$`Away Yellow cards`,
  na.rm = TRUE
)

yellow_card_table <- data.frame(
  YellowCards = yellow_values,
  Home = as.numeric(table(factor(spfl$`Home Yellow cards`,
                                 levels = yellow_values))),
  Away = as.numeric(table(factor(spfl$`Away Yellow cards`,
                                 levels = yellow_values)))
)

yellow_card_table
```

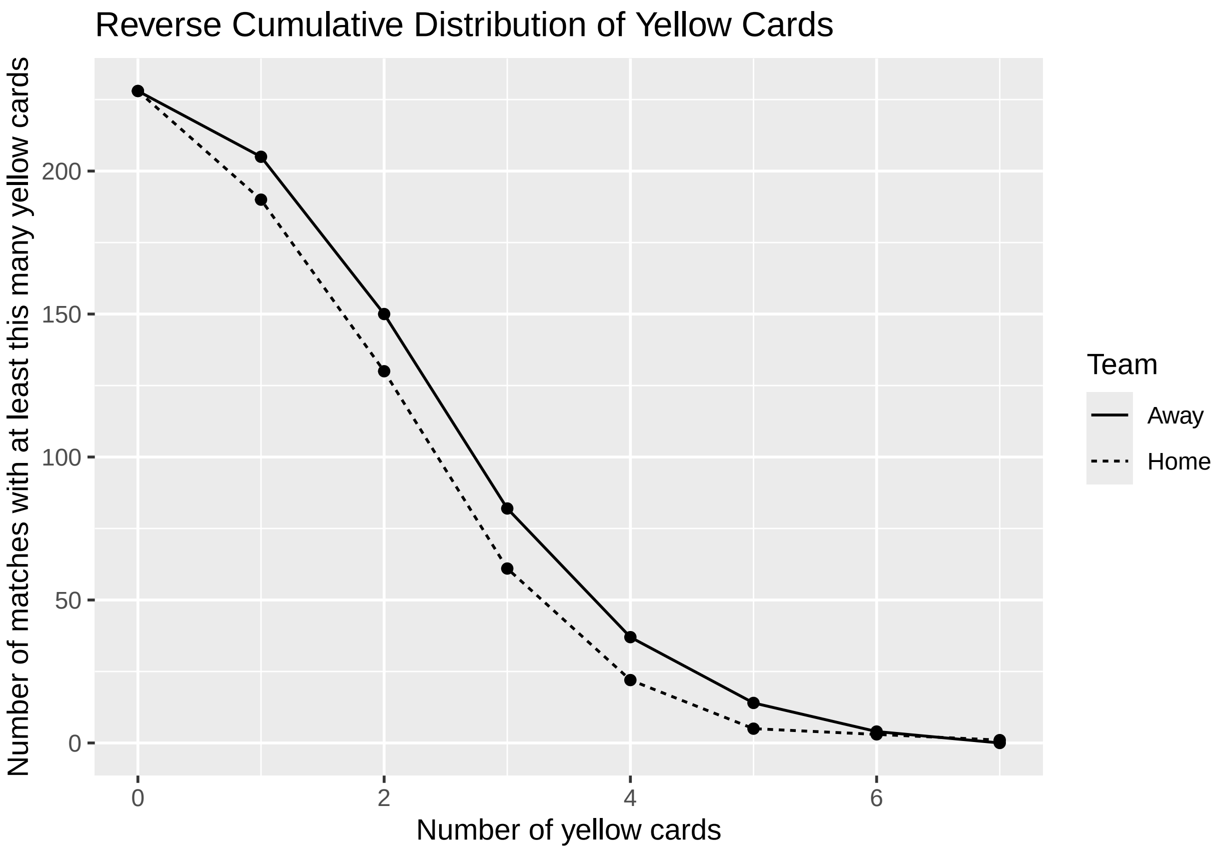
```
##   YellowCards Home Away
## 1           0   38   23
## 2           1   60   55
## 3           2   69   68
## 4           3   39   45
```

```
## 5            4   17   23
## 6            5    2   10
## 7            6    2    4
## 8            7    1    0
```

```r
# Reverse cumulative totals:
# number of matches with at least this many yellow cards
yellow_reverse_cumulative <- yellow_card_table %>%
  arrange(desc(YellowCards)) %>%
  mutate(
    HomeAtLeast = cumsum(Home),
    AwayAtLeast = cumsum(Away)
  ) %>%
  arrange(YellowCards)

yellow_reverse_cumulative
```

```
##   YellowCards Home Away HomeAtLeast AwayAtLeast
## 1           0   38   23         228         228
## 2           1   60   55         190         205
## 3           2   69   68         130         150
## 4           3   39   45          61          82
## 5           4   17   23          22          37
## 6           5    2   10           5          14
## 7           6    2    4           3           4
## 8           7    1    0           1           0
```

```r
yellow_reverse_long <- data.frame(
  YellowCards = rep(yellow_reverse_cumulative$YellowCards, 2),
  Team = c(rep("Home", nrow(yellow_reverse_cumulative)),
           rep("Away", nrow(yellow_reverse_cumulative))),
  MatchesAtLeast = c(yellow_reverse_cumulative$HomeAtLeast,
                     yellow_reverse_cumulative$AwayAtLeast)
)

ggplot(yellow_reverse_long,
       aes(x = YellowCards,
           y = MatchesAtLeast,
           group = Team,
           linetype = Team)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Reverse Cumulative Distribution of Yellow Cards",
    x = "Number of yellow cards",
    y = "Number of matches with at least this many yellow cards",
    linetype = "Team"
  )
```



Red cards

```
home_reds <- table(spfl$`Home Red cards`)
away_reds <- table(spfl$`Away Red cards`)

home_reds
```

```
##
##   0   1   2
## 209  17   2
```

```
away_reds
```

```
##
##   0   1
## 212  16
```

```
t.test(spfl$`Away Red cards`,
       spfl$`Home Red cards`,
       paired = TRUE,
       alternative = "greater")
```

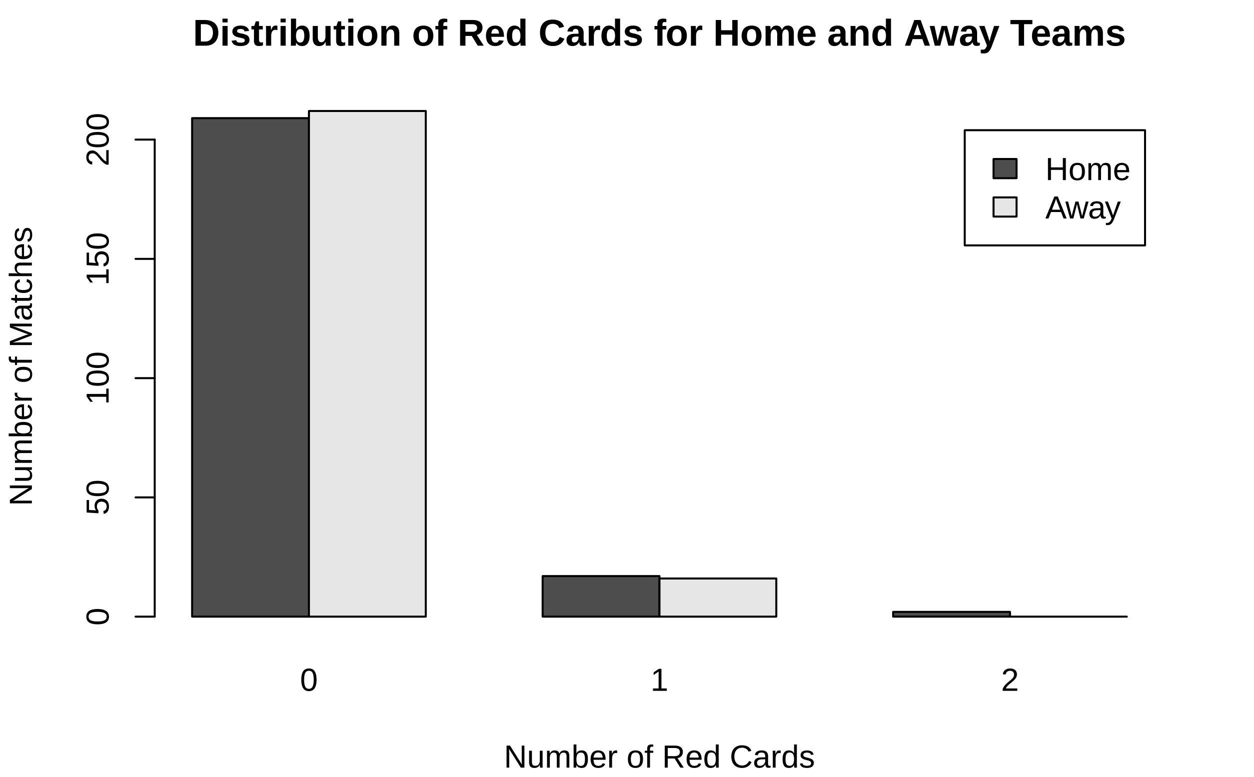
```
##
##  Paired t-test
##
## data:  spfl$`Away Red cards` and spfl$`Home Red cards`
## t = -0.80001, df = 227, p-value = 0.7877
## alternative hypothesis: true mean difference is greater than 0
```

```
## 95 percent confidence interval:
##  -0.0672033        Inf
## sample estimates:
## mean difference
##     -0.02192982
```

The p-value was 0.7877, so the null hypothesis cannot be rejected. There is insufficient evidence that away teams received more red cards than home teams. Red cards occur much less frequently than yellow cards, so this test should be interpreted cautiously.

Visualisation of red cards

```r
red_values <- 0:2

home_reds <- table(factor(spfl$`Home Red cards`, levels = red_values))
away_reds <- table(factor(spfl$`Away Red cards`, levels = red_values))

barplot(
  rbind(home_reds, away_reds),
  beside = TRUE,
  legend.text = c("Home", "Away"),
  main = "Distribution of Red Cards for Home and Away Teams",
  xlab = "Number of Red Cards",
  ylab = "Number of Matches"
)
```



Team-specific card analysis

The following tests compare total cards received at home and away for the top three teams in the league. These team-specific tests should be interpreted cautiously because they use smaller samples and involve multiple comparisons.

Rangers

```
team <- "Rangers"

home_cards <- spfl$`Home Yellow cards`[spfl$`Home Team` == team] +
  spfl$`Home Red cards`[spfl$`Home Team` == team]

away_cards <- spfl$`Away Yellow cards`[spfl$`Away Team` == team] +
  spfl$`Away Red cards`[spfl$`Away Team` == team]

mean(home_cards, na.rm = TRUE)
```

```
## [1] 1.473684
```

```
mean(away_cards, na.rm = TRUE)
```

```
## [1] 2.421053
```

```
t.test(away_cards, home_cards, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  away_cards and home_cards
## t = 2.3545, df = 35.597, p-value = 0.01209
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.2678565       Inf
## sample estimates:
## mean of x mean of y
##  2.421053  1.473684
```

For Rangers, the p-value was 0.01209, which is below 0.05. Therefore, there is statistically significant evidence that Rangers received more cards away from home than at home.

Celtic

```
team <- "Celtic"

home_cards <- spfl$`Home Yellow cards`[spfl$`Home Team` == team] +
  spfl$`Home Red cards`[spfl$`Home Team` == team]

away_cards <- spfl$`Away Yellow cards`[spfl$`Away Team` == team] +
  spfl$`Away Red cards`[spfl$`Away Team` == team]

mean(home_cards, na.rm = TRUE)
```

```
## [1] 2
```

```r
mean(away_cards, na.rm = TRUE)
```

```
## [1] 2.210526
```

```r
t.test(away_cards, home_cards, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  away_cards and home_cards
## t = 0.37966, df = 35.621, p-value = 0.3532
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  -0.7259113        Inf
## sample estimates:
## mean of x mean of y
##  2.210526  2.000000
```

For Celtic, the p-value was 0.3532, which is greater than 0.05. Therefore, there is insufficient evidence that Celtic received more cards away from home than at home.

Hearts

```r
team <- "Hearts"

home_cards <- spfl$`Home Yellow cards`[spfl$`Home Team` == team] +
  spfl$`Home Red cards`[spfl$`Home Team` == team]

away_cards <- spfl$`Away Yellow cards`[spfl$`Away Team` == team] +
  spfl$`Away Red cards`[spfl$`Away Team` == team]

mean(home_cards, na.rm = TRUE)
```

```
## [1] 1.578947
```

```r
mean(away_cards, na.rm = TRUE)
```

```
## [1] 2.315789
```

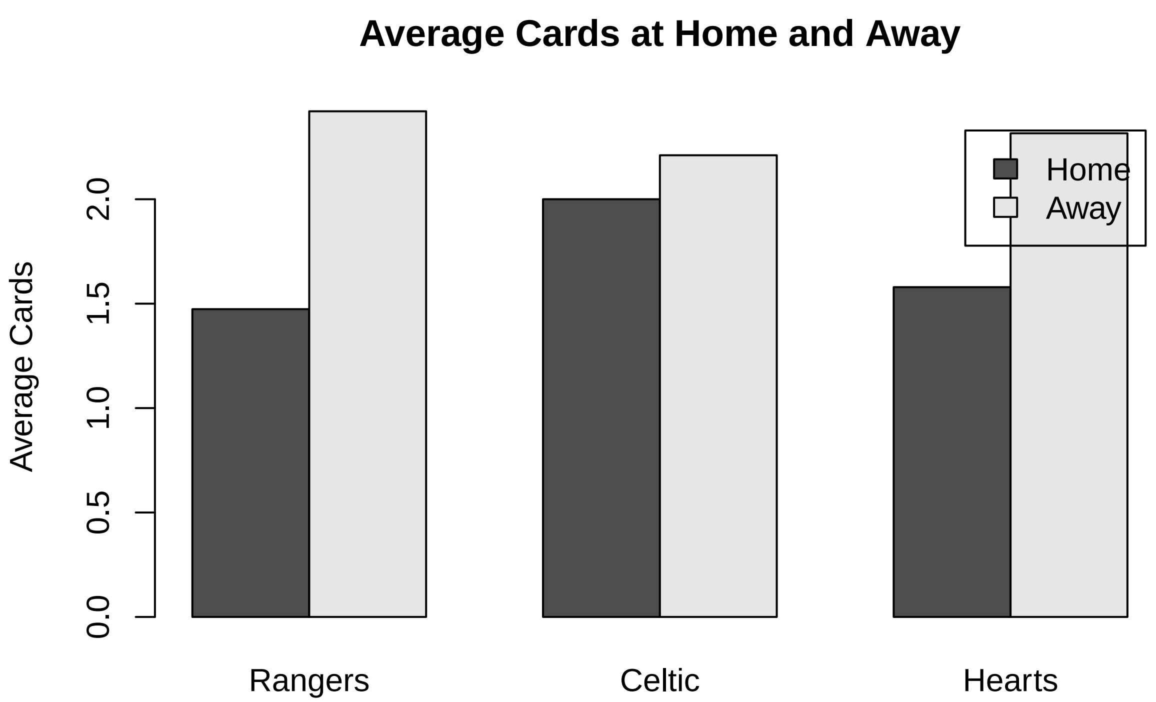
```r
t.test(away_cards, home_cards, alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  away_cards and home_cards
## t = 1.9519, df = 35.692, p-value = 0.02941
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.09936645        Inf
## sample estimates:
## mean of x mean of y
##  2.315789  1.578947
```

For Hearts, the p-value was 0.02941, which is below 0.05. Therefore, there is statistically significant evidence that Hearts received more cards away from home than at home.

Visualisation of team-specific card averages

```r
teams <- c("Rangers", "Celtic", "Hearts")

home_avg <- c(
  mean(spfl$`Home Yellow cards`[spfl$`Home Team` == "Rangers"] + spfl$`Home Red cards`[spfl$`Home Team`
  mean(spfl$`Home Yellow cards`[spfl$`Home Team` == "Celtic"] + spfl$`Home Red cards`[spfl$`Home Team` =
  mean(spfl$`Home Yellow cards`[spfl$`Home Team` == "Hearts"] + spfl$`Home Red cards`[spfl$`Home Team` =
)

away_avg <- c(
  mean(spfl$`Away Yellow cards`[spfl$`Away Team` == "Rangers"] + spfl$`Away Red cards`[spfl$`Away Team`
  mean(spfl$`Away Yellow cards`[spfl$`Away Team` == "Celtic"] + spfl$`Away Red cards`[spfl$`Away Team` =
  mean(spfl$`Away Yellow cards`[spfl$`Away Team` == "Hearts"] + spfl$`Away Red cards`[spfl$`Away Team` =
)

barplot(
  rbind(home_avg, away_avg),
  beside = TRUE,
  names.arg = teams,
  legend.text = c("Home", "Away"),
  main = "Average Cards at Home and Away",
  ylab = "Average Cards"
)
```



Comparing 2025/26 with the 2020/21 COVID season

The 2020/21 season was affected by COVID restrictions, meaning crowds were absent or heavily restricted for much of the season. This section compares disciplinary differences between the two seasons.

Total card difference

The card difference is defined as:

$$\text{Card Difference} = \text{Away Cards} - \text{Home Cards}$$

```r
# Reload full datasets because card columns are needed
spfl <- read_excel("SPFL 2025-2026.xlsx", sheet = "SPFL Dataset")
spfl_2020 <- read_excel("SPFL 2020-2021.xlsx", sheet = "SPFL 2020-2021")

# 2025/26 total card difference
# Positive value means the away team received more cards than the home team
spfl <- spfl %>%
  mutate(
    CardDifference = (`Away Yellow cards` + `Away Red cards`) -
                     (`Home Yellow cards` + `Home Red cards`)
  )

# 2020/21 total card difference
spfl_2020 <- spfl_2020 %>%
  mutate(
    CardDifference = (`Away Yellow cards` + `Away Red cards`) -
                     (`Home Yellow cards` + `Home Red cards`)
  )

# Mean card difference for each season
mean(spfl$CardDifference, na.rm = TRUE)
```

```
## [1] 0.3289474
```

```r
mean(spfl_2020$CardDifference, na.rm = TRUE)
```

```
## [1] -0.004385965
```

```r
t.test(spfl$CardDifference,
       spfl_2020$CardDifference,
       alternative = "greater")
```

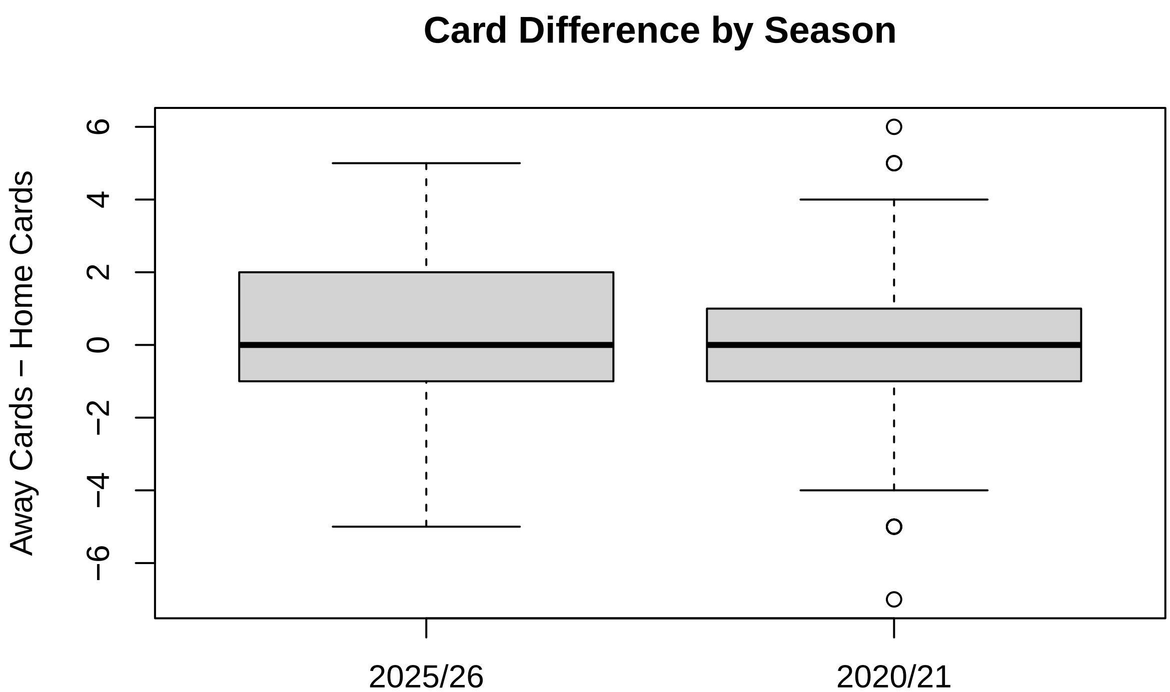
```
##
##  Welch Two Sample t-test
##
## data:  spfl$CardDifference and spfl_2020$CardDifference
## t = 1.8628, df = 452.7, p-value = 0.03157
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.03838882        Inf
## sample estimates:
##    mean of x    mean of y
##  0.328947368 -0.004385965
```

The p-value was 0.03157. In 2025/26, away teams received about 0.33 more cards than home teams per match on average. In 2020/21, the difference was almost zero. This suggests that the disciplinary gap between away and home teams was larger in 2025/26. A positive value means away teams received more cards than home teams.

There is statistically significant evidence that away teams received more cards relative to home teams in the 2025/26 season than in the 2020/21 season. This may suggest that the presence of crowds is associated with a larger disciplinary gap between home and away teams, although other differences between seasons may also have contributed.

Visualisation of card difference by season

```r
boxplot(
  spfl$CardDifference,
  spfl_2020$CardDifference,
  names = c("2025/26", "2020/21"),
  main = "Card Difference by Season",
  ylab = "Away Cards - Home Cards"
)
```



Rangers home performance: 2025/26 compared with 2020/21

Rangers had the highest percent capacity in 2025/26. This section compares Rangers' home performance in 2025/26 with their home performance during the 2020/21 COVID-affected season. The comparison is exploratory, since differences may reflect wider changes in team strength, opposition strength, squad quality, and season context.

```r
rangers_2025_home <- spfl[spfl$`Home Team` == "Rangers", ]
rangers_2020_home <- spfl_2020[spfl_2020$`Home Team` == "Rangers", ]
```

Rangers home win rate

```r
rangers_table <- matrix(
  c(
    sum(rangers_2020_home$Result == "Home"),
    sum(rangers_2020_home$Result != "Home"),
    sum(rangers_2025_home$Result == "Home"),
    sum(rangers_2025_home$Result != "Home")
```

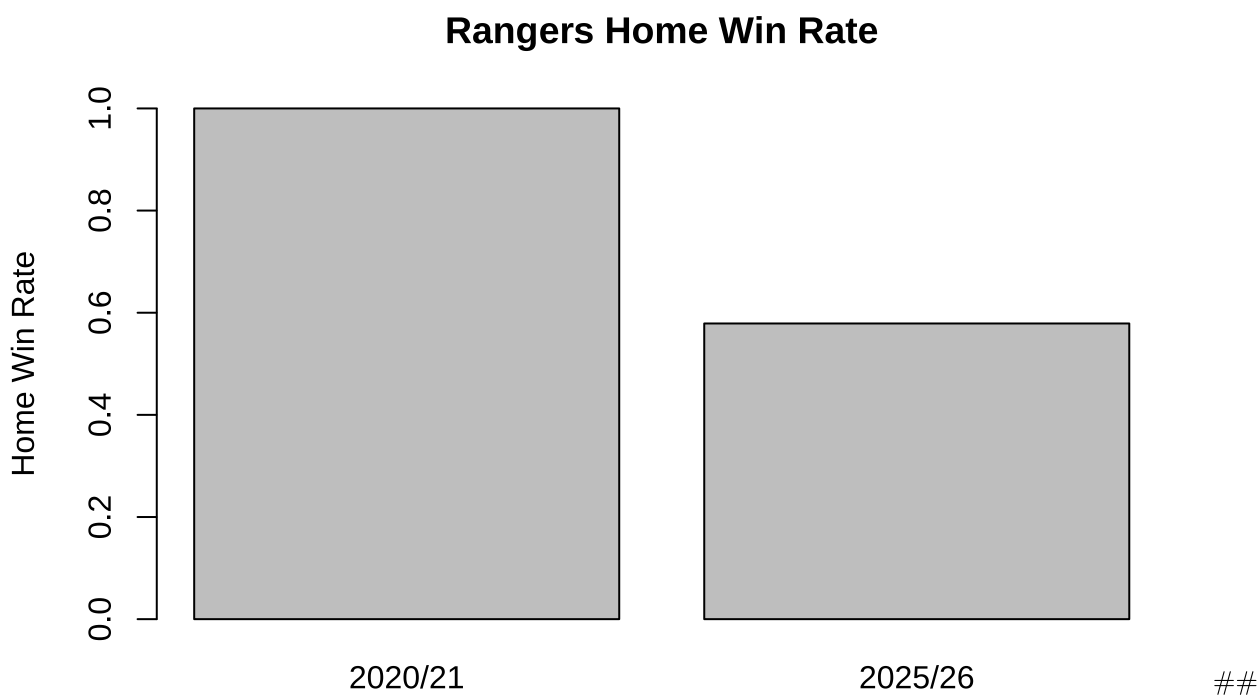
```r
  ),
  nrow = 2,
  byrow = TRUE
)

fisher.test(rangers_table, alternative = "greater")
```

```
##
##  Fisher's Exact Test for Count Data
##
## data:  rangers_table
## p-value = 0.001546
## alternative hypothesis: true odds ratio is greater than 1
## 95 percent confidence interval:
##  3.046621      Inf
## sample estimates:
## odds ratio
##        Inf
```

Fisher's Exact Test gave a p-value of 0.001546, suggesting statistically significant evidence that Rangers' home win rate was higher in 2020/21 than in 2025/26. However, this should not be interpreted as evidence that empty stadiums improved Rangers' performance, because the two seasons differed in many other ways.

Visualisation of Rangers home win rate

```r
rangers_2020_wins <- sum(rangers_2020_home$Result == "Home")
rangers_2025_wins <- sum(rangers_2025_home$Result == "Home")

rangers_2020_games <- nrow(rangers_2020_home)
rangers_2025_games <- nrow(rangers_2025_home)

rangers_win_rates <- c(
  rangers_2020_wins / rangers_2020_games,
  rangers_2025_wins / rangers_2025_games
)

barplot(
  rangers_win_rates,
  names.arg = c("2020/21", "2025/26"),
  main = "Rangers Home Win Rate",
  ylab = "Home Win Rate",
  ylim = c(0, 1)
)
```



Rangers home goal difference

Home goal difference is defined as:

$$\text{Home Goal Difference} = \text{Home Goals} - \text{Away Goals}$$

```
rangers_2025_home$HomeGoalDifference <- rangers_2025_home$`Home Goals` - rangers_2025_home$`Away Goals`
rangers_2020_home$HomeGoalDifference <- rangers_2020_home$`Home Goals` - rangers_2020_home$`Away Goals`

mean(rangers_2025_home$HomeGoalDifference, na.rm = TRUE)
```

```
## [1] 0.9473684
```

```
mean(rangers_2020_home$HomeGoalDifference, na.rm = TRUE)
```

```
## [1] 2.789474
```

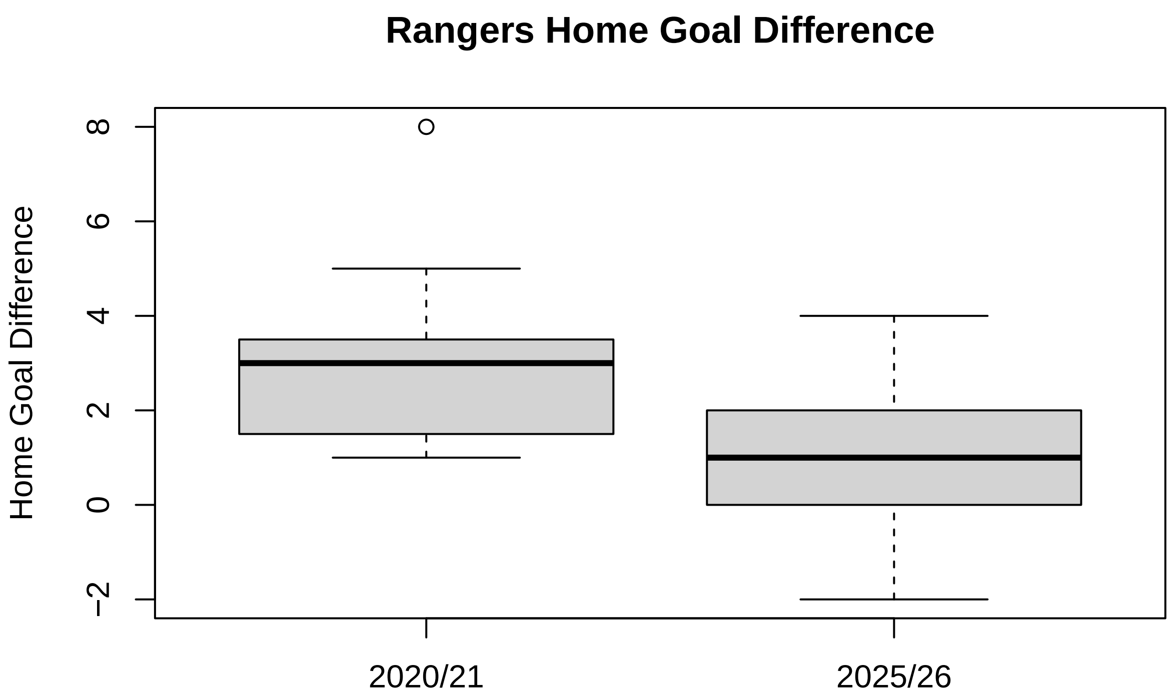
```
t.test(rangers_2020_home$HomeGoalDifference,
       rangers_2025_home$HomeGoalDifference,
       alternative = "greater")
```

```
##
##  Welch Two Sample t-test
##
## data:  rangers_2020_home$HomeGoalDifference and rangers_2025_home$HomeGoalDifference
## t = 3.4394, df = 35.451, p-value = 0.0007542
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.9374985       Inf
## sample estimates:
## mean of x mean of y
## 2.7894737 0.9473684
```

Rangers had a mean home goal difference of 2.79 in 2020/21, compared with 0.95 in 2025/26. The Welch two-sample t-test gave a p-value of 0.0007542, suggesting statistically significant evidence that Rangers' average home goal difference was higher in 2020/21. However, this may reflect wider differences in team strength, opposition strength, squad quality, or season context, rather than crowd effects alone.

Visualisation of Rangers home goal difference

```
boxplot(
  rangers_2020_home$HomeGoalDifference,
  rangers_2025_home$HomeGoalDifference,
  names = c("2020/21", "2025/26"),
  main = "Rangers Home Goal Difference",
  ylab = "Home Goal Difference"
)
```



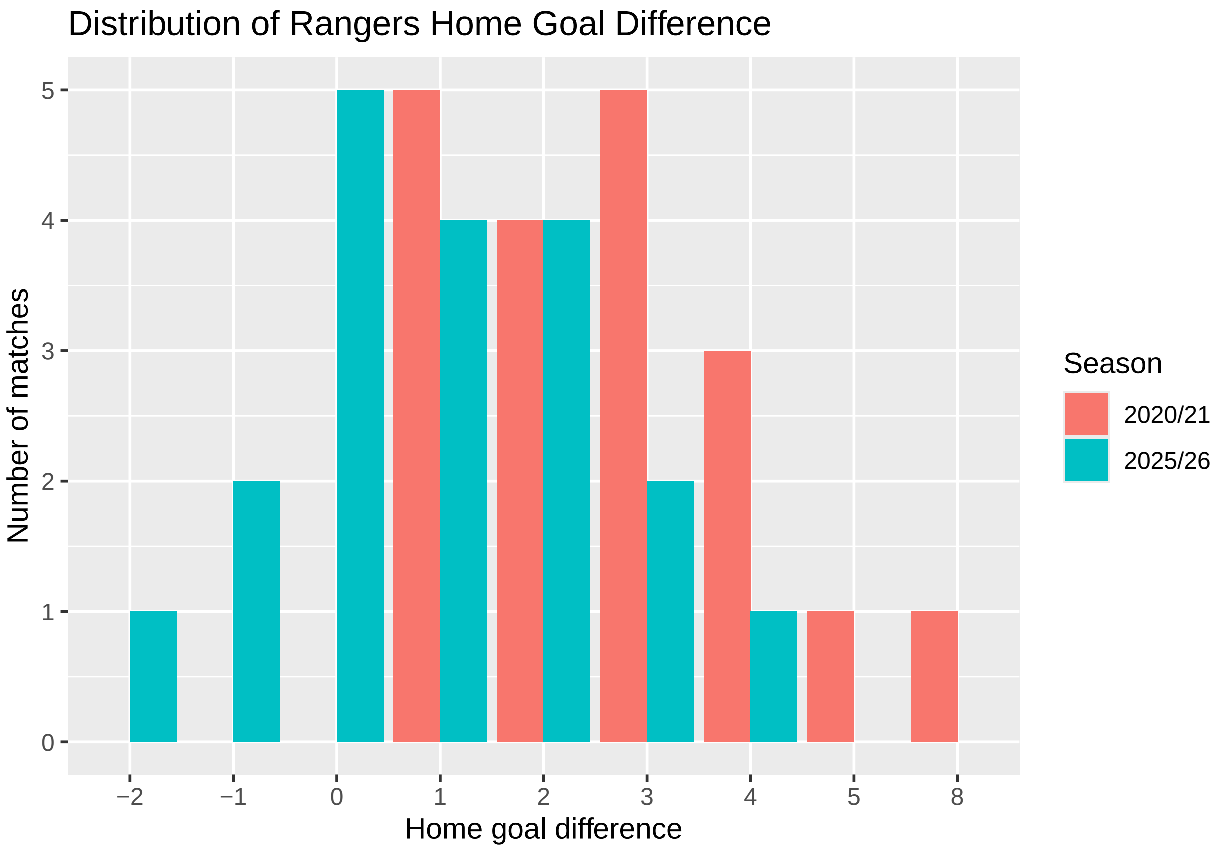

```
# Home goal difference for Rangers home matches in each season
rangers_2025_home <- rangers_2025_home %>%
  mutate(
    HomeGoalDifference = `Home Goals` - `Away Goals`,
    Season = "2025/26"
  )

rangers_2020_home <- rangers_2020_home %>%
  mutate(
    HomeGoalDifference = `Home Goals` - `Away Goals`,
    Season = "2020/21"
  )

# Combine both seasons
rangers_home_gd <- bind_rows(
  rangers_2020_home %>% select(Season, HomeGoalDifference),
  rangers_2025_home %>% select(Season, HomeGoalDifference)
)
```

```r
# Frequency table
rangers_home_gd_table <- rangers_home_gd %>%
  group_by(Season, HomeGoalDifference) %>%
  summarise(
    NumberOfMatches = n(),
    .groups = "drop"
  ) %>%
  arrange(Season, HomeGoalDifference)

rangers_home_gd_table
```

```
## # A tibble: 13 x 3
##    Season  HomeGoalDifference NumberOfMatches
##    <chr>                <dbl>           <int>
##  1 2020/21                  1               5
##  2 2020/21                  2               4
##  3 2020/21                  3               5
##  4 2020/21                  4               3
##  5 2020/21                  5               1
##  6 2020/21                  8               1
##  7 2025/26                 -2               1
##  8 2025/26                 -1               2
##  9 2025/26                  0               5
## 10 2025/26                  1               4
## 11 2025/26                  2               4
## 12 2025/26                  3               2
## 13 2025/26                  4               1
```

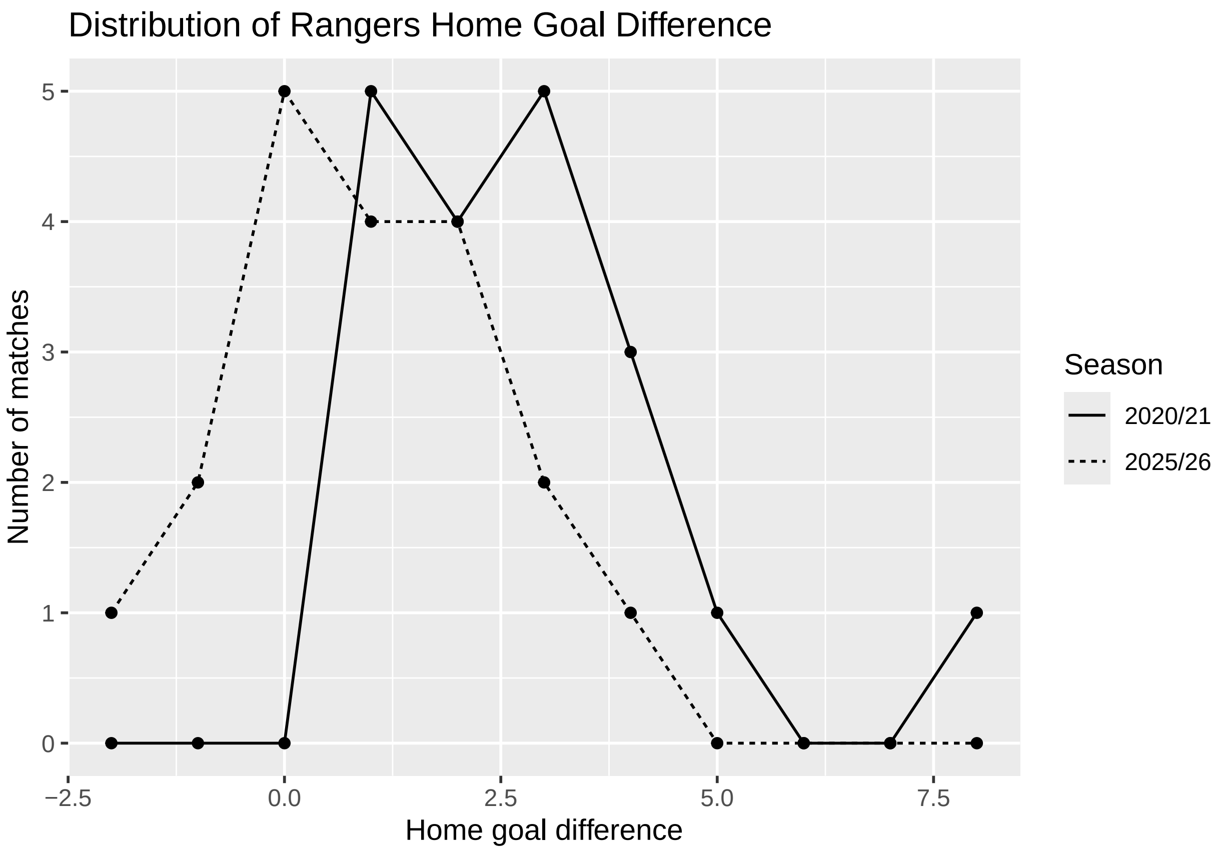
```r
# Make sure both seasons have all goal-difference values included,
# even if one season had 0 matches at that value.

all_goal_differences <- sort(unique(rangers_home_gd_table$HomeGoalDifference))

full_rangers_gd_table <- expand.grid(
  Season = c("2020/21", "2025/26"),
  HomeGoalDifference = all_goal_differences
) %>%
  left_join(
    rangers_home_gd_table,
    by = c("Season", "HomeGoalDifference")
  ) %>%
  mutate(
    NumberOfMatches = ifelse(is.na(NumberOfMatches), 0, NumberOfMatches)
  )

ggplot(full_rangers_gd_table,
       aes(x = factor(HomeGoalDifference),
           y = NumberOfMatches,
           fill = Season)) +
  geom_col(position = "dodge") +
  labs(
    title = "Distribution of Rangers Home Goal Difference",
    x = "Home goal difference",
    y = "Number of matches",
```

```
    fill = "Season"
  )
```




```
# Make sure both seasons have all goal-difference values included,
# even if one season had 0 matches at that value.

all_goal_differences <- seq(
  min(rangers_home_gd_table$HomeGoalDifference),
  max(rangers_home_gd_table$HomeGoalDifference),
  by = 1
)

full_rangers_gd_table <- expand.grid(
  Season = c("2020/21", "2025/26"),
  HomeGoalDifference = all_goal_differences
) %>%
  left_join(
    rangers_home_gd_table,
    by = c("Season", "HomeGoalDifference")
  ) %>%
  mutate(
    NumberOfMatches = ifelse(is.na(NumberOfMatches), 0, NumberOfMatches)
  )

ggplot(full_rangers_gd_table,
       aes(x = HomeGoalDifference,
           y = NumberOfMatches,
```

```
           group = Season,
           linetype = Season)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Distribution of Rangers Home Goal Difference",
    x = "Home goal difference",
    y = "Number of matches",
    linetype = "Season"
  )
```



#

Home and Away wins from 2016-2026 comparison

```
library(readxl)
library(dplyr)
library(ggplot2)

# Function to load one season and keep only the Result column
load_results <- function(file, season, sheet = NULL) {

  if (is.null(sheet)) {
    data <- read_excel(file)
  } else {
    data <- read_excel(file, sheet = sheet)
  }

  data %>%
    select(Result) %>%
```

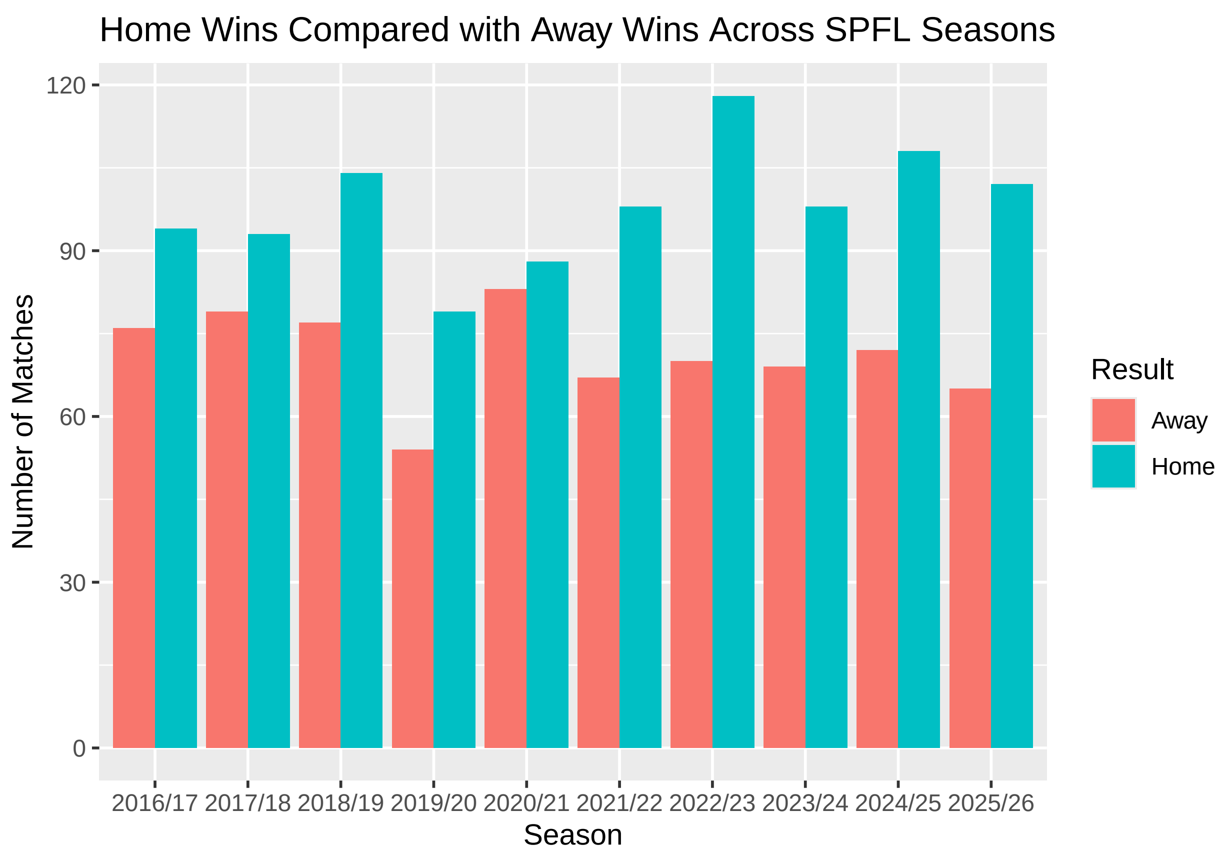
```
    mutate(Season = season)
}

# Load each season
spfl_2016_2017 <- load_results("SPFL 2016-2017.xlsx", "2016/17", "SPFL 2016-2017")
spfl_2017_2018 <- load_results("SPFL 2017-2018.xlsx", "2017/18", "SPFL 2017-2018")
spfl_2018_2019 <- load_results("SPFL 2018-2019.xlsx", "2018/19", "SPFL 2018-2019")
spfl_2019_2020 <- load_results("SPFL 2019-2020.xlsx", "2019/20", "SPFL 2019-2020")
spfl_2020_2021 <- load_results("SPFL 2020-2021.xlsx", "2020/21", "SPFL 2020-2021")
spfl_2021_2022 <- load_results("SPFL 2021-2022.xlsx", "2021/22")
spfl_2022_2023 <- load_results("SPFL 2022-2023.xlsx", "2022/23")
spfl_2023_2024 <- load_results("SPFL 2023-2024.xlsx", "2023/24")
spfl_2024_2025 <- load_results("SPFL 2024-2025.xlsx", "2024/25")
spfl_2025_2026 <- load_results("SPFL 2025-2026.xlsx", "2025/26", "SPFL Dataset")

# Combine all seasons into one dataset
all_seasons_results <- bind_rows(
  spfl_2016_2017,
  spfl_2017_2018,
  spfl_2018_2019,
  spfl_2019_2020,
  spfl_2020_2021,
  spfl_2021_2022,
  spfl_2022_2023,
  spfl_2023_2024,
  spfl_2024_2025,
  spfl_2025_2026
)

all_seasons_results
```

```
## # A tibble: 2,231 x 2
##    Result Season
##    <chr>  <chr>
##  1 Away   2016/17
##  2 Home   2016/17
##  3 Draw   2016/17
##  4 Away   2016/17
##  5 Away   2016/17
##  6 Draw   2016/17
##  7 Draw   2016/17
##  8 Away   2016/17
##  9 Away   2016/17
## 10 Away   2016/17
## # i 2,221 more rows
```

```
# Count home wins and away wins only, excluding draws
home_away_wins <- all_seasons_results %>%
  filter(Result %in% c("Home", "Away")) %>%
  group_by(Season, Result) %>%
  summarise(NumberOfMatches = n(), .groups = "drop")

home_away_wins
```

```
## # A tibble: 20 x 3
##    Season  Result NumberOfMatches
##    <chr>   <chr>            <int>
##  1 2016/17 Away                76
##  2 2016/17 Home                94
##  3 2017/18 Away                79
##  4 2017/18 Home                93
##  5 2018/19 Away                77
##  6 2018/19 Home               104
##  7 2019/20 Away                54
##  8 2019/20 Home                79
##  9 2020/21 Away                83
## 10 2020/21 Home                88
## 11 2021/22 Away                67
## 12 2021/22 Home                98
## 13 2022/23 Away                70
## 14 2022/23 Home               118
## 15 2023/24 Away                69
## 16 2023/24 Home                98
## 17 2024/25 Away                72
## 18 2024/25 Home               108
## 19 2025/26 Away                65
## 20 2025/26 Home               102
```

```r
# Bar plot
ggplot(home_away_wins,
       aes(x = Season, y = NumberOfMatches, fill = Result)) +
  geom_col(position = "dodge") +
  labs(
    title = "Home Wins Compared with Away Wins Across SPFL Seasons",
    x = "Season",
    y = "Number of Matches",
    fill = "Result"
  )
```




```r
home_advantage_summary <- home_away_wins %>%
  group_by(Season) %>%
  summarise(
    HomeWins = NumberOfMatches[Result == "Home"],
    AwayWins = NumberOfMatches[Result == "Away"],
    TotalDecidedGames = HomeWins + AwayWins,
    HomeWinPercentage = HomeWins / TotalDecidedGames,
    Difference = HomeWins - AwayWins,
    .groups = "drop"
  )

home_advantage_summary
```

```
## # A tibble: 10 x 6
##    Season  HomeWins AwayWins TotalDecidedGames HomeWinPercentage Difference
##    <chr>      <int>    <int>             <int>             <dbl>      <int>
##  1 2016/17       94       76               170             0.553         18
##  2 2017/18       93       79               172             0.541         14
##  3 2018/19      104       77               181             0.575         27
##  4 2019/20       79       54               133             0.594         25
##  5 2020/21       88       83               171             0.515          5
##  6 2021/22       98       67               165             0.594         31
##  7 2022/23      118       70               188             0.628         48
##  8 2023/24       98       69               167             0.587         29
##  9 2024/25      108       72               180             0.6           36
## 10 2025/26      102       65               167             0.611         37
```

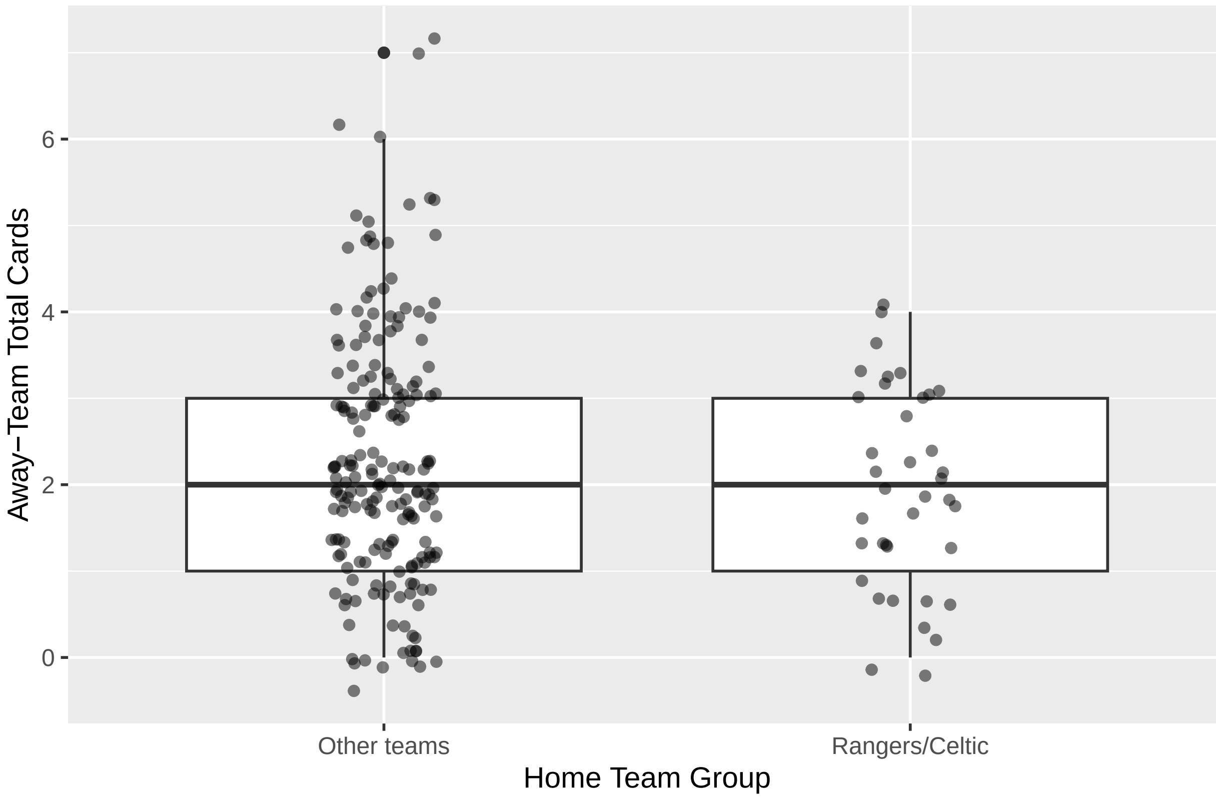
```r
# Do teams receive more cards away at Rangers/Celtc?
```

```r
spfl$BigClub <- ifelse(spfl$`Home Team` %in% c("Rangers", "Celtic"),
                       "Rangers/Celtic",
                       "Other teams")

spfl$AwayTotalCards <- spfl$`Away Yellow cards` + spfl$`Away Red cards`

t.test(AwayTotalCards ~ BigClub, data = spfl)
```

```
##
##  Welch Two Sample t-test
##
## data:  AwayTotalCards by BigClub
## t = 1.749, df = 64.448, p-value = 0.08506
## alternative hypothesis: true difference in means between group Other teams and group Rangers/Celtic i
## 95 percent confidence interval:
##  -0.05234242  0.78918452
## sample estimates:
##    mean in group Other teams mean in group Rangers/Celtic
##                     2.289474                     1.921053
```

```r
ggplot(spfl, aes(x = BigClub, y = AwayTotalCards)) +
  geom_boxplot() +
  geom_jitter(width = 0.1, alpha = 0.5) +
  labs(
    title = "Away-Team Cards at Rangers/Celtic Compared with Other Teams",
    x = "Home Team Group",
    y = "Away-Team Total Cards"
  )
```

Away−Team Cards at Rangers/Celtic Compared with Other Teams



Key Match incident(KMI) data for 2024-25, 2025-26

```r
# Load packages needed for data manipulation and plotting.
library(dplyr)
library(ggplot2)

# Create a dataset of incorrect Key Match Incident decisions for 2024/25.
# DecisionsFor records incorrect decisions that benefited each team.
# DecisionsAgainst records incorrect decisions that went against each team.
kmi_2024_2025 <- data.frame(
  Season = "2024/25",
  Team = c("Celtic", "Rangers", "Hibernian", "Dundee United",
           "Aberdeen", "St Mirren", "Hearts", "Motherwell",
           "Kilmarnock", "Dundee", "Ross County", "St Johnstone"),
  DecisionsFor = c(3, 2, 4, 3, 1, 4, 1, 2, 1, 1, 3, 3),
  DecisionsAgainst = c(2, 5, 0, 2, 1, 5, 4, 3, 4, 0, 2, 0)
)

# Create the equivalent KMI dataset for 2025/26.
kmi_2025_2026 <- data.frame(
  Season = "2025/26",
  Team = c("Celtic", "Hearts", "Rangers", "Motherwell",
           "Hibernian", "Falkirk", "Dundee United", "Dundee",
           "Aberdeen", "Kilmarnock", "St Mirren", "Livingston"),
  DecisionsFor = c(2, 1, 1, 2, 1, 3, 0, 0, 0, 2, 2, 3),
```

```r
  DecisionsAgainst = c(1, 1, 3, 4, 1, 0, 1, 1, 1, 0, 3, 1)
)

# Combine both seasons into one dataset so that the KMI data
# can be analysed across the two seasons together.
kmi_all <- bind_rows(kmi_2024_2025, kmi_2025_2026)
```

```r
# Summarise the combined KMI data by team.
# This gives the total number of incorrect decisions for and against
# each team across the 2024/25 and 2025/26 seasons.
kmi_all_teams <- kmi_all %>%
  group_by(Team) %>%
  summarise(
    TotalFor = sum(DecisionsFor, na.rm = TRUE),
    TotalAgainst = sum(DecisionsAgainst, na.rm = TRUE),
    NetDifferential = TotalFor - TotalAgainst,
    .groups = "drop"
  ) %>%

  # Order teams from the highest to lowest net differential.
  arrange(desc(NetDifferential))

# Display the team-level summary table.
kmi_all_teams
```

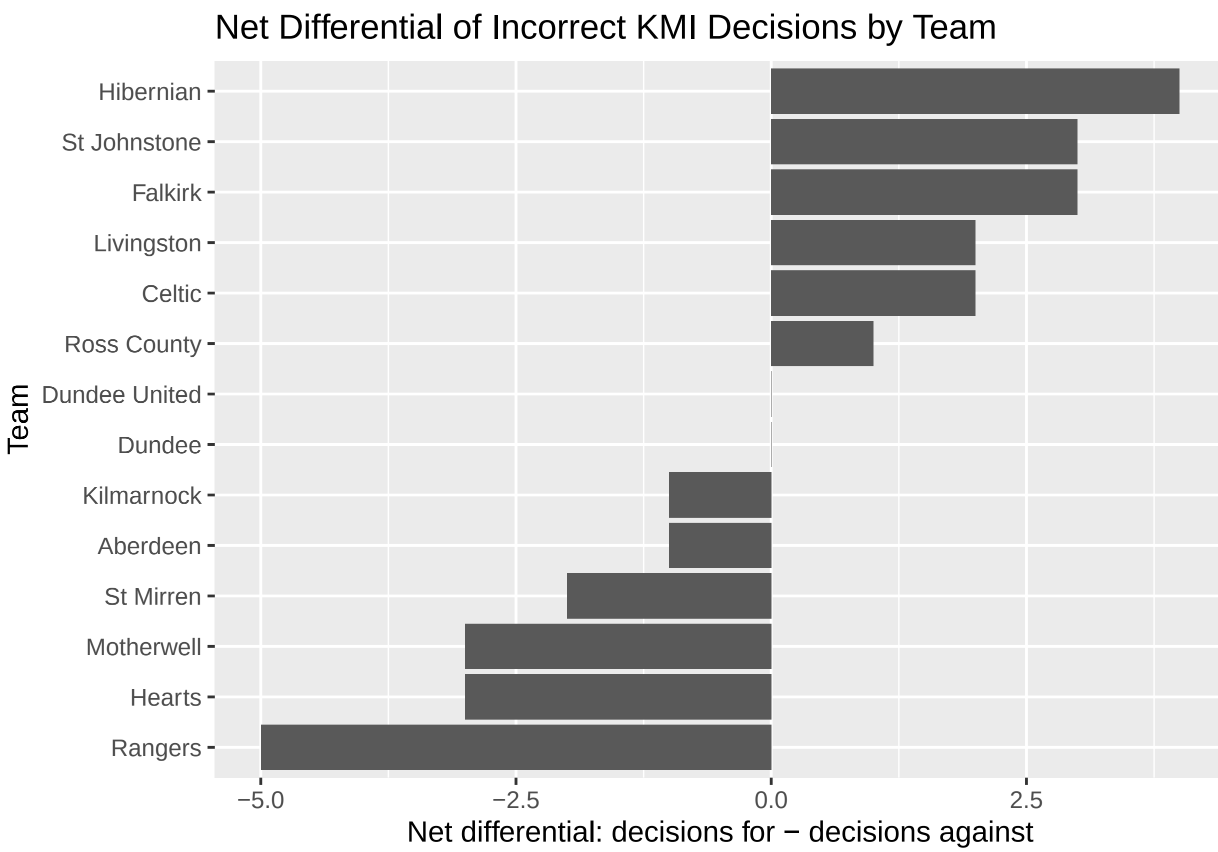
```
## # A tibble: 14 x 4
##    Team          TotalFor TotalAgainst NetDifferential
##    <chr>            <dbl>        <dbl>           <dbl>
##  1 Hibernian            5            1               4
##  2 Falkirk              3            0               3
##  3 St Johnstone         3            0               3
##  4 Celtic               5            3               2
##  5 Livingston           3            1               2
##  6 Ross County          3            2               1
##  7 Dundee               1            1               0
##  8 Dundee United        3            3               0
##  9 Aberdeen             1            2              -1
## 10 Kilmarnock           3            4              -1
## 11 St Mirren            6            8              -2
## 12 Hearts               2            5              -3
## 13 Motherwell           4            7              -3
## 14 Rangers              3            8              -5
```

```r
# Carry out a chi-squared goodness-of-fit test on the distribution
# of incorrect decisions against teams.
chisq.test(kmi_all_teams$TotalAgainst)
```

```
##
##  Chi-squared test for given probabilities
##
## data:  kmi_all_teams$TotalAgainst
## X-squared = 31.844, df = 13, p-value = 0.002533
```

```r
# Produce a bar chart showing the net differential of incorrect KMI decisions
# for each team across the combined 2024/25 and 2025/26 data.
```

```r
ggplot(kmi_all_teams,
       aes(x = reorder(Team, NetDifferential),
           y = NetDifferential)) +

  # Add bars representing each team's net differential.
  geom_col() +

  # Flip the axes so that team names are easier to read.
  coord_flip() +

  # Add plot labels.
  labs(
    title = "Net Differential of Incorrect KMI Decisions by Team",
    x = "Team",
    y = "Net differential: decisions for - decisions against"
  )
```




```r
team_tests <- data.frame()

for (team_name in kmi_all_teams$Team) {

  team_for <- kmi_all_teams$TotalFor[kmi_all_teams$Team == team_name]
  team_against <- kmi_all_teams$TotalAgainst[kmi_all_teams$Team == team_name]

  other_for <- sum(kmi_all_teams$TotalFor[kmi_all_teams$Team != team_name])
  other_against <- sum(kmi_all_teams$TotalAgainst[kmi_all_teams$Team != team_name])
```

```r
  test_table <- matrix(
    c(team_against, team_for,
      other_against, other_for),
    nrow = 2,
    byrow = TRUE
  )

  rownames(test_table) <- c(team_name, "Other teams")
  colnames(test_table) <- c("Against", "For")

  test <- fisher.test(test_table, alternative = "greater")

  team_tests <- rbind(
    team_tests,
    data.frame(
      Team = team_name,
      TotalFor = team_for,
      TotalAgainst = team_against,
      NetDifferential = team_for - team_against,
      PValue = test$p.value
    )
  )
}

# Adjust p-values because several teams are tested
team_tests <- team_tests %>%
  mutate(
    AdjustedPValue = p.adjust(PValue, method = "holm")
  ) %>%
  arrange(PValue)

team_tests
```

```
##              Team TotalFor TotalAgainst NetDifferential     PValue
## 1         Rangers        3            8              -5 0.09829925
## 2          Hearts        2            5              -3 0.21702145
## 3      Motherwell        4            7              -3 0.26081304
## 4       St Mirren        6            8              -2 0.38608272
## 5        Aberdeen        1            2              -1 0.50000000
## 6      Kilmarnock        3            4              -1 0.50000000
## 7   Dundee United        3            3               0 0.66170203
## 8          Dundee        1            1               0 0.75280899
## 9     Ross County        3            2               1 0.81964355
## 10         Celtic        5            3               2 0.86684931
## 11     Livingston        3            1               2 0.94168927
## 12      Hibernian        5            1               4 0.98691798
## 13        Falkirk        3            0               3 1.00000000
## 14   St Johnstone        3            0               3 1.00000000
##    AdjustedPValue
## 1               1
## 2               1
## 3               1
## 4               1
## 5               1
```

```
## 6               1
## 7               1
## 8               1
## 9               1
## 10              1
## 11              1
## 12              1
## 13              1
## 14              1
```

```r
# Create team-level totals from the combined KMI dataset.
# This calculates the total number of incorrect decisions for and against
# each team across the two seasons.
kmi_all_teams <- kmi_all %>%
  group_by(Team) %>%
  summarise(
    TotalFor = sum(DecisionsFor, na.rm = TRUE),
    TotalAgainst = sum(DecisionsAgainst, na.rm = TRUE),
    TotalDecisions = TotalFor + TotalAgainst,
    NetDifferential = TotalFor - TotalAgainst,
    .groups = "drop"
  )

# Carry out an exact binomial test for each team.
# The test checks whether the proportion of incorrect decisions against
# a team differs significantly from 50% of all incorrect decisions involving
# that team.
kmi_binomial_tests <- kmi_all_teams %>%
  rowwise() %>%
  mutate(

    # TotalAgainst is treated as the number of "successes",
    # while TotalDecisions is the total number of incorrect decisions
    # for and against that team.
    PValue = binom.test(
      x = TotalAgainst,
      n = TotalDecisions,
      p = 0.5,
      alternative = "two.sided"
    )$p.value
  ) %>%
  ungroup() %>%

  # Adjust the p-values using the Holm method to account for
  # multiple team-level tests.
  mutate(
    AdjustedPValue = p.adjust(PValue, method = "holm")
  ) %>%

  # Order the results from smallest to largest unadjusted p-value.
  arrange(PValue)

# Display the binomial test results.
kmi_binomial_tests
```

```
## # A tibble: 14 x 7
##    Team           TotalFor TotalAgainst TotalDecisions NetDifferential PValue
##    <chr>             <dbl>        <dbl>          <dbl>           <dbl>  <dbl>
##  1 Hibernian             5            1              6               4  0.219
##  2 Rangers               3            8             11              -5  0.227
##  3 Falkirk               3            0              3               3  0.25
##  4 St Johnstone          3            0              3               3  0.25
##  5 Hearts                2            5              7              -3  0.453
##  6 Motherwell            4            7             11              -3  0.549
##  7 Livingston            3            1              4               2  0.625
##  8 Celtic                5            3              8               2  0.727
##  9 St Mirren             6            8             14              -2  0.791
## 10 Aberdeen              1            2              3              -1  1
## 11 Dundee                1            1              2               0  1
## 12 Dundee United         3            3              6               0  1
## 13 Kilmarnock            3            4              7              -1  1
## 14 Ross County           3            2              5               1  1
## # i 1 more variable: AdjustedPValue <dbl>
```

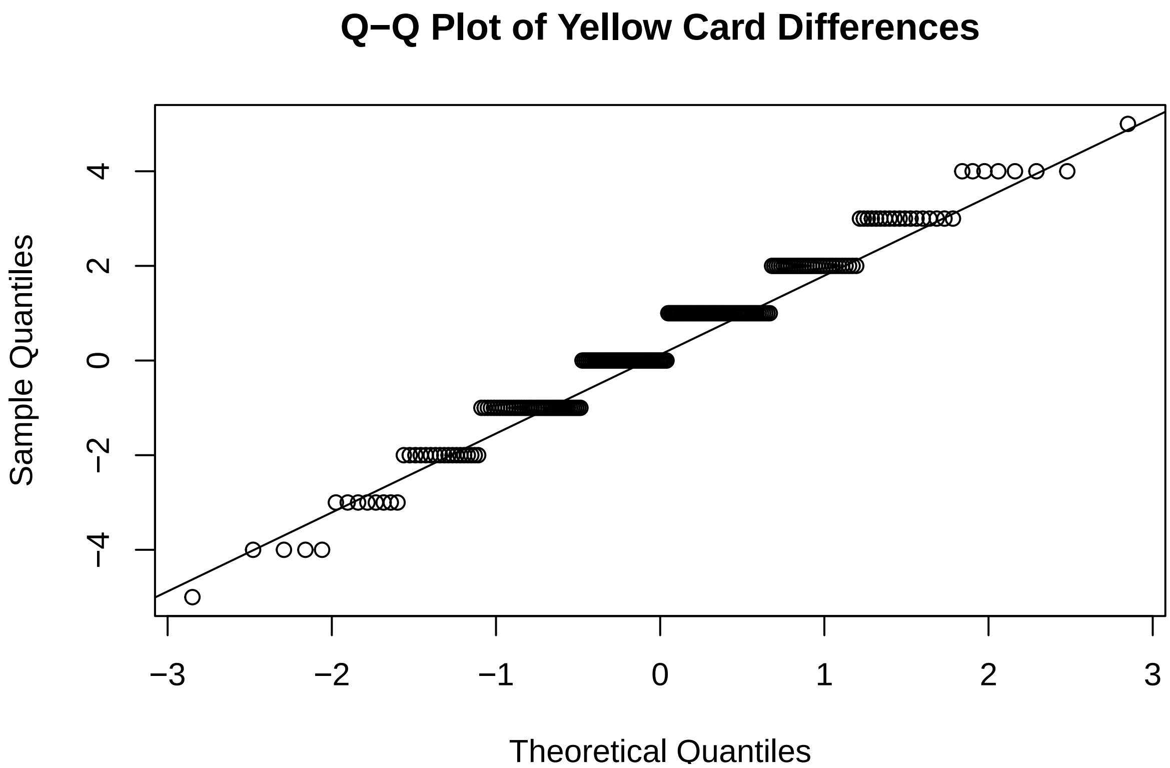
```r
kmi_binomial_against_tests <- kmi_all_teams %>%
  rowwise() %>%
  mutate(
    PValue = binom.test(
      x = TotalAgainst,
      n = TotalDecisions,
      p = 0.5,
      alternative = "greater"
    )$p.value
  ) %>%
  ungroup() %>%
  mutate(
    AdjustedPValue = p.adjust(PValue, method = "holm")
  ) %>%
  arrange(PValue)

kmi_binomial_against_tests
```

```
## # A tibble: 14 x 7
##    Team           TotalFor TotalAgainst TotalDecisions NetDifferential PValue
##    <chr>             <dbl>        <dbl>          <dbl>           <dbl>  <dbl>
##  1 Rangers               3            8             11              -5  0.113
##  2 Hearts                2            5              7              -3  0.227
##  3 Motherwell            4            7             11              -3  0.274
##  4 St Mirren             6            8             14              -2  0.395
##  5 Kilmarnock            3            4              7              -1  0.5
##  6 Aberdeen              1            2              3              -1  0.5
##  7 Dundee United         3            3              6               0  0.656
##  8 Dundee                1            1              2               0  0.75
##  9 Ross County           3            2              5               1  0.812
## 10 Celtic                5            3              8               2  0.855
## 11 Livingston            3            1              4               2  0.938
## 12 Hibernian             5            1              6               4  0.984
## 13 Falkirk               3            0              3               3  1
## 14 St Johnstone          3            0              3               3  1
## # i 1 more variable: AdjustedPValue <dbl>
```

Additional visuals/tests

Q-Q plot for yellow-card differences

```r
# Difference in yellow cards: away minus home
spfl$YellowCardDifference <- spfl$`Away Yellow cards` - spfl$`Home Yellow cards`

# Q-Q plot
qqnorm(spfl$YellowCardDifference,
       main = "Q-Q Plot of Yellow Card Differences")

qqline(spfl$YellowCardDifference)
```



Wilcoxon signed-rank test for yellow cards

```r
wilcox.test(spfl$`Away Yellow cards`,
            spfl$`Home Yellow cards`,
            paired = TRUE,
            alternative = "greater")
```

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  spfl$`Away Yellow cards` and spfl$`Home Yellow cards`
## V = 10430, p-value = 0.001283
## alternative hypothesis: true location shift is greater than 0
```